

CHEMISTRY

EQUATIONS & ANSWERS™

Essential Tool for Chemistry Concepts, Variables & Equations, Including Sample Problems, Common Pitfalls & Helpful Hints

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Your success in chemistry depends on your ability to solve numerical problems.

BASIC SKILLS

CALCULATOR SURVIVAL

- Become familiar with your calculator **before** the exam; make sure you can multiply, divide, add, subtract and use all needed functions.
 - Calculators never make mistakes; they take your input (intended & accidental) and give an answer.
 - Look at the answer; does it make sense?
 - Do a quick estimate to check your work.
- Sample:** $4.34 \times 7.68 / (1.05 \times 9.8)$ is roughly $4 \times 8 / (1 \times 10)$ or 3.2; the actual answer: **3.24**.

HOW TO DO WORD PROBLEMS

- Read and evaluate the question **before** you start plugging numbers into the calculator.
- Identify the variables, constants and equations.
- Write out units of the variables and constants.
- Work out the unit before the number.
- You may have extra information, or you may need to obtain constants from your text.

HOW TO WORK WITH UNITS

All numerical data has units. In chemistry, we use metric "SI" units.

Pitfall: If the unit is wrong, the answer is wrong!

1. Unit prefixes: Denote powers of "10"

tera T 10^{12}	giga G 10^9	mega M 10^6
kilo k 10^3	deci d 10^{-1}	centi c 10^{-2}
milli m 10^{-3}	micro μ 10^{-6}	nano n 10^{-9}
pico p 10^{-12}	femto f 10^{-15}	

2. Fundamental Units

- Mass:** kilogram (kg)
- Length:** meter (m)
- Temperature:** Kelvin (K)
- Time:** second (s)
- Amount** of a substance: mole
- Electric charge:** coulomb, (C)

3. Derived Units

- Area:** length squared, m^2
- Volume:** length cubed, m^3 ; 1 liter (L) = $1 dm^3$
- Density:** mass/volume; common unit kg/m^3
- Speed:** distance/time; common unit m/s
- Electric current:** ampere (A) = $1 C/s$
- Force:** Newton (N) = $1 kg \cdot m/s^2$
- Energy:** Joule (J) = $1 kg \cdot m^2/s^2$
- Pressure:** Pascal (Pa) = $1 kg/(m \cdot s^2)$

4. Fundamental Constants

Mass	in amu (g/mole)	in kg
electron:	5.486×10^{-4}	9.10939×10^{-31}
proton:	1.007276	1.67262×10^{-27}
neutron:	1.008664	1.67493×10^{-27}
Electronic charge: $1.6022 \times 10^{-19} C$		
Avogadro's number, N_A: 6.02214×10^{23}		
Ideal gas constant, R:		
R, for gas calculation: $0.082 L \cdot atm/(K \cdot mol)$		
R, for energy calculation: $8.314510 J/(K \cdot mol)$		
Faraday Constant, $\mathcal{F}$:	96,485 C/mol	
Planck's Constant, h:	$6.626 \times 10^{-34} J \cdot s$	
Speed of Light, c:	$2.9979 \times 10^8 m/s$	

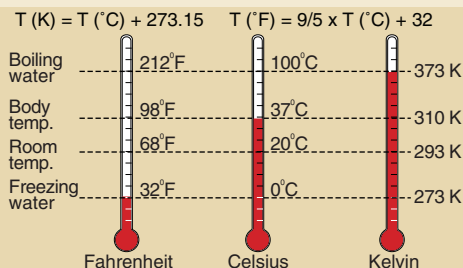
BASIC SKILLS cont.

HOW TO CONVERT DATA

The unit & numerical value are **changed** using a conversion factor or equation.

HOW TO USE EQUATIONS FOR DATA CONVERSION

Sample: Convert a temperature of $45^\circ C$ to $^\circ F$ and Kelvin:
Given: $K = ^\circ C + 273.15$
Calculate: Temperature in K = $45^\circ C + 273.15$
 $= 318.15 K$
Given: $^\circ F = 9/5 ^\circ C + 32$
Calculate: Temperature in $^\circ F = 9/5 \times 45 ^\circ C + 32$
 $= 81 + 32 = 113 ^\circ F$



HOW TO USE A CONVERSION FACTOR

Sample: "1 hour = 60 minutes" gives two conversion factors:
1. Multiply by "1 hour/60 minutes" to convert minutes to hours.
2. Multiply by "60 minutes/1 hour" to convert hours to minutes.
Sample: Determine the number of hours in 45 minutes.
Given: Conversion factor is "1 hour/60 minutes."
Calculate: Time = 45 minutes $\times$ (1 hour/60 minutes)
 $= 0.75$ hour (minutes cancel).

Common conversions:

1 calorie = 4.184 J	1 kg = 2.2 lb	1 m = 1.1 yd
1 qt = 0.9464 L	1 angstrom ($\text{\AA}$) = $1 \times 10^{-10} m$	
1 atm = $1.01325 \times 10^5 Pa$	1 atm = 760 mm Hg	

Pitfall: Equations and conversion factors have units. Beware of inverting conversion factors.

MATH REVIEW

HOW TO WORK WITH ALGEBRAIC EQUATIONS

Give equal treatment to each side.

• Add or subtract:

Sample: Given, $x = y$, then, $4 + x = 4 + y$.

• Multiply or divide:

Sample: Given, $x = y$, then, $4x = 4y$ and $x/5 = y/5$.
Given: $a = b + 4$, then $a/(x - 2) = (b + 4)/(x - 2)$, $x \neq 2$.

Pitfall: Dividing by zero is not allowed.

HOW TO WORK WITH SCIENTIFIC NOTATION

The exponent gives the power of 10.

Sample: $0.00045 = 4.5 \times 10^{-4}$; $1345 = 1.345 \times 10^3$
Sample chemical applications: Molecule diameters are $10^{-10} m$; 1 liter of water contains about 1×10^{26} atoms.

MATH REVIEW cont.

HOW TO USE LOGARITHMS & EXPONENTS

- Common logarithm, $\log_{10}$:** base "10."
Denotes number or function in powers of 10.
Sample: Given, $y = 10^6$, then $\log_{10} y = 6$.
 - Natural logarithm, $\ln$:** base "e" ($e = 2.718281829$).
Denotes number or function in powers of "e."
Sample: Given, $z = e^5$, then $\ln z = 5$.
 - Products:** $\log(xy) = \log x + \log y$
 - Powers:** $\log(x^n) = n \log(x)$
 - Multiplication:** Add exponents from each term:
Sample: $10^5 \times 10^3 = 10^{(5+3)} = 10^8$.
 - Division:** Subtract denominator exponents from numerator exponents:
Sample: $10^5/10^3 = 10^{(5-3)} = 10^2$.
 - Square root:** $\sqrt{a} = a^{1/2}$
 - Inverse:** $1/x = x^{-1}$
- Pitfall:** Your calculator has separate keys for $\ln x$ (base e), $\log x$ (base 10), 10^x and e^x .
- Sample chemical applications: pH of acid and base.

HOW TO CALCULATE ROOTS OF A POLYNOMIAL

An equation of the form: $ax^2 + bx + c = 0$ has 2 solutions or roots, given by the quadratic formula:

$$\frac{-b + \sqrt{(b^2 - 4ac)}}{2a} \quad \text{Root 1}$$

$$\frac{-b - \sqrt{(b^2 - 4ac)}}{2a} \quad \text{Root 2}$$

Sample: Determine the roots for the equation:
 $3x^2 + 4x + 1 = 0$.

Given: $a = 3$, $b = 4$, $c = 1$

Calculate: Root 1 = $\frac{-4 + \sqrt{(4^2 - 4 \times 3 \times 1)}}{(2 \times 3)}$
 $= \frac{-4 + \sqrt{(16 - 12)}}{6} = \frac{-4 + 2}{6} = -1/3$

Calculate: Root 2 = $\frac{-4 - \sqrt{(16 - 12)}}{6} = \frac{-4 - 2}{6} = -1$

Pitfall: Beware round-off error. Substitute the roots into the equation to verify results.

Sample chemical applications: Weak acids, weak bases, buffers, chemical equilibrium.

HOW TO DETERMINE THE EQUATION OF A LINE

LINEAR EQUATION: $y = mx + b$

m: slope of the line; $m = \Delta y / \Delta x$

b: y-intercept, the line crosses the y-axis at "b"; $b = y_i - mx_i$

Sample: Determine the equation of a line using (x,y) data.

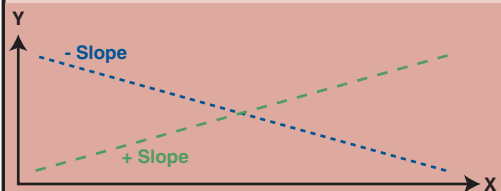
Given: x:	-2	-1	0	1	2	3
y:	-2	1	4	7	10	13

Calculate: Slope = $m = \Delta y / \Delta x$
 $= [13 - (-2)] / [3 - (-2)] = 15/5 = 3$

Calculate: y-intercept = $b = y_i - mx_i$
 $= (-2) - 3 \times (-2) = -2 + 6 = 4$

The equation of the line is $y = 3x + 4$.

Sample chemical applications: Gas calculations; Beer's Law; analyzing reaction-rate data.



STATISTICS

HOW TO CALCULATE EXPERIMENTAL ERROR

- **Accuracy:** The agreement between experimental data and a known value.

- **Error:** A measure of accuracy:
Error = (experiment value - known value).

$$\text{Relative error} = \frac{\text{Error}}{\text{known value}}$$

$$\% \text{ error} = \text{relative error} \times 100\%$$

Sample: A student finds the mass of an object to be 5.51g; the correct value is 5.80g. What is the % error?

Given: experimental value = 5.51g;

known value = 5.80g.

Calculate: error = (exp. value - known value)
= 5.51 - 5.80g = -0.29g

Calculate: relative error = error/known value
= -0.29g/5.8g = -0.050

Calculate: % error = relative error x 100%
= -0.05 x 100% = -5.0%

In this case, we undershoot the known value by 5%.

- **Precision:** The degree to which a set of experimental values agree with each other.

Pitfall: A set of data can be precise, but have a large experimental error.

CALCULATING MEAN & DEVIATION

For a set of numbers, $\{x_1, x_2, x_3 \dots x_j\}$:

- The **mean**, or average value, is the sum of all "x" divided by "j," the # of entries.
- The deviation, Δ_i , for each x_i is $x_i - \text{mean}$; Δ_i can be positive or negative.

Sample: Determine the mean and deviations for the following data: {25, 28, 31, 35, 43, 48}.

Mean = (25 + 28 + 31 + 35 + 43 + 48)/6 = 35

x_i	25	28	31	35	43	48
Δ_i	25-35	28-35	31-35	35-35	43-35	48-35
Δ_i	-10	-7	-4	0	+8	+13

Note: Sum of Δ_i = 0.

HOW TO IDENTIFY SIGNIFICANT FIGURES (sig. figs.)

- Record all certain digits and one uncertain or estimated digit for a measurement.
- Note: A calculator often includes extra digits in a calculation.
- For a multi-step problem, keep 1 or 2 extra digits; then, round off the **final** answer.

Rules for the # of sig. figs. in the final answer:

- For **addition** or subtraction: Use the **least** number of decimal places found in the data.

Sample: 10.102 + 5.03 = 15.13 (2 dec. places).

- For **multiplication** or **division**: The final answer should have same # of sig. figs. as the entry with the **fewest** sig. figs.

Sample: 5.46200 x 4.00 = 21.8 (3 sig. figs.).

- **Rounding-off data:** Round up if the last uncertain digit is 6,7,8,9; round down if it is 0,1,2,3,4. If it is a 5: The **arbitrary** convention is to round up if the last **certain** digit is odd, round down if it is even.

Sample: 0.085 $\Rightarrow$ 0.08; 0.035 $\Rightarrow$ 0.04

0.453 $\Rightarrow$ 0.45 0.248 $\Rightarrow$ 0.25

ATOMIC DATA

- Atomic Number: $Z = \#$ of protons.
- Atomic Mass Number: $A = \#$ of protons + $\#$ of neutrons.
- A and Z are **integers**. The actual particle mass is given in kg or **amu (g/mole)**. The actual mass is **close** in value to A .

HOW TO CALCULATE NUCLEAR BINDING ENERGY

The nuclear mass does not equal the sum of proton and neutron masses. The mass

difference (Δm , kg/mole) is $\Delta E = \Delta m c^2$ due to the nuclear binding

energy (ΔE , J):

Sample: Calculate the binding energy for He-4.

Given: He-4 has 2 protons and 2 neutrons

Given: Proton particle mass: 2 x 1.00728 amu

Given: Neutron particle mass: 2 x 1.00866 amu

Calculate: Total particle mass = Sum of proton and neutron particle mass = 4.03188 amu

Given: Actual mass of He-4 nucleus: 4.00150 amu

Calculate: Δm = Particle mass - He-4 mass
= 0.03038 amu = 0.03038 g/mole

Convert to kg: $\Delta m = 3.038 \times 10^{-5}$ kg/mole

Calculate: $\Delta E = \Delta m c^2$
= 3.038×10^{-5} kg/mole x $(3.00 \times 10^8 \text{ m/s})^2$
= 2.73×10^{12} J/mole

Pitfall: Watch units of mass; you can work in kg/particle or amu (g/mole). Note: J = kg m²/s².

CALCULATING ATOMIC WEIGHT

For an element with **two isotopes**:

Atomic wt = $\text{mass}_a \times \text{fract}_a + \text{mass}_b \times \text{fract}_b$

Sample: Calculate the atomic weight for chlorine.

Given: Chlorine has two isotopes

	mass	fractional abundance
Cl-35	34.968852	0.7577
Cl-37	36.965303	0.2423

Calculate: Atomic wt = $\text{mass}_a \times \text{fract}_a + \text{mass}_b \times \text{fract}_b$
= $34.968852 \times 0.7577 + 36.965303 \times 0.2423$
= 26.496 + 8.9566
= 35.45 amu (4 sig. figs.)

Pitfall: Use the actual mass of the nucleus, not the mass number.

HOW TO CALCULATE PROPERTIES OF ELECTROMAGNETIC RADIATION

Light waves are characterized by **wavelength** (λ , in m), **frequency** (ν , in Hertz, s⁻¹) and the speed of light (c , in m/s). The energy is carried in **photons**.

$$\lambda \nu = c$$

$$\text{Energy of a photon} = h\nu$$

Sample: Calculate the ν and energy for λ of 500 nm.

Given: $\lambda = 500$ nm and $1 \text{ nm} = 1 \times 10^{-9} \text{ m}$

Convert λ to m: $\lambda = 500 \text{ nm} = 5.00 \times 10^{-7} \text{ m}$

Calculate: $\nu = c/\lambda = 3.00 \times 10^8 \text{ m/s} / 5.00 \times 10^{-7} \text{ m}$
= $6.00 \times 10^{14} \text{ s}^{-1}$ (Hertz)

Calculate: energy = $h\nu$
= $6.626 \times 10^{-34} \text{ J s} \times 6.00 \times 10^{14} \text{ s}^{-1}$
= $3.98 \times 10^{-19} \text{ J}$ (s cancels)

Pitfall: The unit on λ should match the unit of c .

CHEMICAL FORMULAS & MOLES

The **formula** and **name** denote elements and relative composition in the compound. A **balanced** equation conserves **atoms** and **moles** of each element.

- One **mole** is Avogadro's number, N_A (6.023×10^{23}), of atoms or molecules.
- The **atomic weight**, the mass in grams, of one mole of the atomic element as found in nature, is often given on the Periodic Table, along with the atomic number and element symbol.
- The **molar mass** of a compound is the mass, in grams, of 1 mole of the substance.

HOW TO CALCULATE MOLAR MASS FROM THE FORMULA

Given: The **atomic weight** (at.wt.) of each element in the compound and the formula coefficients.

Calculate: Molar mass = sum of each element's atomic weight multiplied by the formula coefficient.

Sample: Simple case: MgCl_2 .

Given: Mg at.wt.: 24.305 g/mole; coefficient = 1

Cl at.wt.: 35.453 g/mole; coefficient = 2

Calculate: Molar mass = 1 x Mg at.wt. + 2 x Cl at.wt.
= 1 x 24.305 + 2 x 35.453
= 95.211 g/mole

Sample: Complex case: $\text{Mg}(\text{NO}_3)_2 \times 2\text{H}_2\text{O}$.

Given: Mg at.wt.: 24.305 g/mole; coefficient = 1

N at.wt.: 14.007 g/mole; coefficient = 2

O at.wt.: 15.9994 g/mole; coefficient = 8

H at.wt.: 1.008 g/mole; coefficient = 4

Calculate: Molar mass = 1 x Mg at.wt. + 2 x N at.wt. + 8 x O at.wt. + 4 x H at.wt.
= 24.305 + 2 x 14.007
+ 8 x 15.9994 + 4 x 1.008
= 24.305 + 28.014
+ 127.9952 + 4.032
= 184.35 g/mole

Pitfall: It is easy to miscount the atoms in polyatomic ions or waters of hydration.

HOW TO CALCULATE ELEMENTAL % COMP

The portion of the mass coming from each element in the compound; the %-comps sum to 100%.

Given: Chemical formula, compound molar mass and atomic weights for each element.

Step 1: Sum the formula coefficients to determine the number of atoms of each element.

Step 2: The mass of each element = at.wt. x number of atoms of that element.

Calculate: % comp for element A = 100% x mass of A/compound molar mass

Sample: Determine the elemental % comp of Mg and Cl in MgCl_2 .

Given: MgCl_2 , molar mass = 95.21 g/mole;

Mg at.wt. = 24.305 g/mole;

Cl at.wt. = 35.453 g/mole

Step 1: MgCl_2 has 1 atom of Mg

MgCl_2 has 2 atoms of Cl

Step 2: Mass of Mg = 24.305 g/mole x 1
= 24.305 g/mole
Mass of Cl = 35.453 g/mole x 2
= 70.906 g/mole

Calculate: **Mg % comp**
= 100% x 24.305/95.21
= 25.53% Mg

Calculate: **Cl % comp**
= 100% x 70.906/95.21 = 74.47 %

Sum of elemental % comp
= 25.53% + 74.47% = 100.00%

Pitfall: In a complex formula, the same element may exist in different ions.

CHEMICAL FORMULAS & MOLES *cont.*

HOW TO CALCULATE THE # OF MOLECULES

Given: x , the mass of the sample, and the molar mass of the material

Calculate: Number of moles = $\frac{x \text{ (in g)}}{\text{molar mass (g/mole)}}$

Calculate: Number of molecules = # of moles $\times$ Avogadro's number (N_A)

Sample: Determine the number of moles and the number of water molecules in 5.00 grams of water vapor.

Given: 5.00 g of H_2O ; molar mass = 18.015 g/mole

Calculate: # of moles = $5.00g H_2O / 18.015g/mole H_2O = 0.278 \text{ moles } H_2O$

Calculate: # of molecules = $0.278 \text{ moles } H_2O \times 6.022 \times 10^{23} \text{ molecules/mole} = 1.67 \times 10^{23} H_2O \text{ molecules}$

This could also be determined in a single calculation:
 $5.00g \times 1 \text{ mole} / 18.015g \times N_A \text{ molecules/mole} = 1.67 \times 10^{23} \text{ molecules}$

HOW TO CALCULATE EMPIRICAL FORMULAS

Given: % composition and atomic weights, start by converting “%” to “grams of element”; assume you have 100 g of sample.

Calculate: Moles for each element = grams of element $\div$ atomic weight of the element.

Calculate: The formula coefficient for each element = number of moles $\div$ the smaller number of moles calculated in the previous step.

To determine the **empirical formula**, multiply each coefficient by a number to give whole # coefficient.

Sample: Determine the empirical formula for a compound containing 75% C & 25 % H by mass.

Given: Atomic weights: C at.wt. = 12.011;

H at.wt. = 1.008.

Assume you have 75 g of C and 25 g of H (total mass = 100 g).

Calculate: The number of moles for each element:
 $75 \text{ g of C} / 12.011g/mole \text{ C} = 6.24 \text{ moles of C}$

$25g \text{ of H} / 1.008 \text{ g/mole H} = 24.8 \text{ moles of H.}$

Calculate: The formula coefficient for each element:
 C formula coefficient: $6.24/6.24 = 1.00$.
 H formula coefficient: $24.8/6.24 = 3.97$.
 The empirical formula is CH_4 .

Pitfall: Due to experimental error, the calculated formula coefficients may not be integers.

HOW TO DERIVE THE MOLECULAR FORMULA FROM THE EMPIRICAL FORMULA

The molecular formula coefficients are obtained by multiplying each empirical formula coefficient by a **factor**: The molecular molar mass $\div$ empirical molar mass.

Given: The empirical formula, molecular molar mass, and atomic weights.

Calculate: Empirical molar mass = sum of at.wts. in the empirical formula.

Calculate: The molecular $\div$ empirical factor = molecular molar mass $\div$ empirical molar mass.
 Determine the molecular formula by multiplying each empirical coefficient by this factor.

Sample: Determine the molecular formula for a compound with molar mass 28 and empirical formula CH_2 .

Given: The empirical formula is CH_2 ;

C at.wt. = 12.0; H at.wt. = 1.0.

Calculate: The empirical molar mass = $12 + 2 = 14$.

Calculate: The molecular/empirical factor = $28/14 = 2$.
 The **molecular** formula is C_2H_4 ; the coefficients are **twice** the **empirical** formula coefficients.

STOICHIOMETRY

HOW TO USE BALANCED EQUATIONS

Sample: $2 \text{ Mg} + O_2 \Rightarrow 2 \text{ MgO}$.

2 Mg atoms and 2 O atoms on each side

2 moles of Mg and 2 moles of O on each side

Calculate the masses of the product and reactants:

Given: A balanced equation and atomic weights:

$Mg = 24.31$; $O = 16.00$.

Calculate: Mass of $Mg = 2 \times Mg \text{ at.wt.} = 48.62g \text{ of } Mg$.

Calculate: Mass of $O_2 = 2 \times O \text{ at.wt.} = 32.00g \text{ of } O_2$.

Calculate: Mass of $MgO = 2 \times Mg \text{ at.wt.} + 2 \times O \text{ at.wt.} = 80.62g \text{ of } MgO$.

$48.62g \text{ of } Mg \text{ reacts with } 32.00g \text{ of } O_2 \text{ to produce } 80.62g \text{ of } MgO$.

Balanced equation conserves mass; $80.62g$ on each side.

HOW TO BALANCE AN EQUATION

First: Identify each **element** in the reaction.

Next: Determine the net charge on each side; this must be balanced in the final equation.

Guidelines: Start with the element found in one compound on each side. Identify compounds that must have the same coefficient. If an element appears in a pure form, leave it to the last step.

You should always check your work!

Pitfall: Coefficients apply to **each** atom in a molecule or polyatomic ion. You can change coefficients in the equation, **not** the formula subscripts.

Sample: $CH_4 + O_2 \Rightarrow CO_2 + H_2O$

First: Elements: C, H, O.

Next: No charge to worry about.

1. H is in CH_4 and H_2O ; start with H.

2. The H_2O coefficient must be twice the CH_4 coefficient to balance “H.”

$\bullet 1 CH_4 + 2 O_2 \Rightarrow 2 CO_2 + 2 H_2O$

3. CH_4 and CO_2 must have the same coefficient.

$\bullet 1 CH_4 + 2 O_2 \Rightarrow 1 CO_2 + 2 H_2O$

4. Now, determine the O_2 coefficient:

$\bullet 1 CH_4 + 2 O_2 \Rightarrow 1 CO_2 + 2 H_2O$

5. Check your work: 1 C, 4 H and 4 O on each side...It is balanced!

BALANCING A REDOX EQUATION USING THE HALF-REACTION METHOD

$\bullet$ Split the reaction into **oxidation** and **reduction** half-reactions.

$\bullet$ You may need to add H_2O and H^+ for acidic, or H_2O and OH^- for basic reaction conditions.

$\bullet$ Balance these separately, then combine to balance the exchange of electrons.

Sample: Balance the following for acidic solution:
 $MnO_4^- + Fe^{2+} \Rightarrow Mn^{2+} + Fe^{3+}$.

1. In acidic solution: Add H^+ to the left and H_2O to the right side:

$MnO_4^- + Fe^{2+} + H^+ \Rightarrow H_2O + Mn^{2+} + Fe^{3+}$

2. Identify the half-reactions:

$Fe^{2+} \Rightarrow Fe^{3+}$ (oxidation)

$MnO_4^- + H^+ \Rightarrow Mn^{2+} + H_2O$ (reduction)

3. Add electrons to account for valence changes:

$Fe^{2+} \Rightarrow Fe^{3+} + 1e^-$ Fe(II) to Fe(III)

$5e^- + MnO_4^- + H^+ \Rightarrow Mn^{2+} + H_2O$ Mn(VII) to Mn(II)

4. Balance each half-reaction:

a. **Oxidation:** Multiply by a factor of 5 to match electrons in reduction step:

$5 Fe^{2+} \Rightarrow 5 Fe^{3+} + 5 e^-$

Charge: +10 on each side, balanced!

b. **Reduction:** Balance O, then H^+ ;

check charge: $5e^- + MnO_4^- + 8H^+ \Rightarrow Mn^{2+} + 4H_2O$
 Charge: +2 on each side, balanced!

5. Combine half-reactions to eliminate the $5e^-$:

$5 Fe^{2+} + MnO_4^- + 8 H^+ \Rightarrow Mn^{2+} + 4 H_2O + 5 Fe^{3+}$

Check your work: 5 Fe, 1 Mn, 4 O and 8 H on each side, atoms are balanced!

Charge: +17 on each side, charge is balanced!

Pitfall: Make sure you use the H_2O , H^+ (for acidic), or OH^- (for basic), with the correct half-reaction.

STOICHIOMETRY

HOW TO CALCULATE THE THEORETICAL YIELD

Mass of a **reactant** is used to determine mass of **product**.

Given: Mass of reactant, balanced equation, molar masses of reactants and products

Calculate: Moles of reactant = $\frac{\text{reactant mass}}{\text{reactant molar mass}}$

Calculate: *Molar ratio* = $\frac{\text{product equation coefficient}}{\text{reactant equation coefficient}}$

Calculate: Moles of product

= moles of reactant $\times$ *molar ratio*

Calculate: Mass of product

= moles of product $\times$ product molar mass

Sample: Calculate the mass of Mg produced by burning 10.0g of Mg in excess oxygen.

Balanced equation: $2 \text{ Mg} + O_2 \Rightarrow 2 \text{ MgO}$

Given: 10.0 grams of Mg , $Mg \text{ at.wt.} = 24.305 \text{ g}$,
 $MgO \text{ molar mass} = 40.305 \text{ g}$

Calculate: Moles of Mg

= $10.0g \text{ of } Mg / 24.305g \text{ Mg/mole Mg}$

= 0.411 moles of Mg

Calculate: Molar ratio = $2/2 = 1$

Calculate: Moles of MgO

= 0.411 moles $Mg \times 1 \text{ mole } MgO/\text{mole } Mg$

= 0.411 moles of MgO

Calculate: Mass of $MgO = 0.411 \text{ moles } MgO$

$\times 40.305 \text{ g } MgO/\text{mole } MgO$

= 16.6 g MgO

The calculation can also be performed as a single sequence of calculations:

$10.0g \text{ Mg} \times 1 \text{ mol Mg} \times 1 \text{ mol MgO} \times 40.35g \text{ MgO}$
 $24.305g \text{ Mg} \quad 1 \text{ mol Mg} \quad 1 \text{ mol MgO}$
 = 16.6 g MgO (note canceled units in *italics*)

Pitfall: If your balanced equation is wrong, your theoretical yield will usually be wrong.

HOW TO CALCULATE A LIMITING REAGENT FOR 2 REACTANTS

In a reaction with 2 reactants, the mass of product is constrained by the reactant in shortest supply, the **limiting reagent**.

Given: Balanced equation, mass of reactants, molar masses of reactants; specify reactant #s.

Calculate: Moles of each reactant =

$\frac{\text{mass}}{\text{molar mass}}$

Calculate: *Ideal reactant molar ratio* = coefficient of reactant #1/coefficient of reactant #2

Calculate: *Actual reactant molar ratio* = moles of reactant #1/moles of reactant #2

Determine the limiting reagent:

$\bullet$ If *actual reactant molar ratio* $\leq$ *ideal reactant molar ratio*, then reactant #1 is the limiting reagent.

$\bullet$ If *actual reactant molar ratio* $>$ *ideal reactant molar ratio*, then reactant #2 is the limiting reagent.

Calculate the **theoretical yield** based on the **mass of the limiting reagent**.

Hint: The reactant numbering is arbitrary, but you must stick with your choice for the entire calculation.

Sample: 10.0g Mg reacts with 10.0g O_2 ; how much MgO is produced?

Balanced equation: $2 \text{ Mg} + O_2 \Rightarrow 2 \text{ MgO}$

Given: $Mg \text{ molar mass} = 24.305g$; Reactant #1

$O_2 \text{ molar mass} = 32.00g$; Reactant #2

Calculate: Moles of $Mg = 10.0g \text{ Mg} / 24.305g$

$Mg/\text{mol Mg} = 0.411 \text{ mol Mg}$

moles of $O_2 = 10.0g \text{ } O_2 / 32.00g \text{ } O_2/\text{mol}$

$O_2 = 0.3125 \text{ mol } O_2$

Calculate: Ideal reactant molar ratio = $2/1 = 2$

Calculate: Actual reactant molar ratio

= moles $Mg/\text{moles } O_2 = 0.411/0.3125 = 1.31$

Determine limiting reagent: 1.31 is less than 2.0, therefore, Mg is the limiting reagent.

Calculate the yield of MgO based on 10.0 grams of Mg (shown in the previous section).

Pitfall: Make sure you distinguish between the ideal and actual molar ratios.

WORKING WITH GASES

Simple model for gas behavior.

$$PV = nRT$$

Ideal Gas Law

Pressure, P; common units: atm, Pa, bar or mm Hg; the R given below is for P in atm.

Volume, V; common units: liter (L), m³; the R given below is for V in L.

Temperature, T; common units: Kelvin, °C or °F; always convert °C and °F to Kelvin.

Number of moles, n; moles = gas mass/gas molar mass

Ideal Gas constant = R = 0.082 L atm/(mol K)

Pitfall: All data must fit the units of R.

HOW TO CALCULATE THE # OF MOLES OF A GAS SAMPLE

Given: Mass of gas (g), molar mass of gas

Calculate: Moles = mass of sample/molar mass

Sample: Determine # of moles in 5.0g of H₂ gas.

Given: 5.0g sample, H₂ molar mass = 2.016 g/mole.

Calculate: Moles of H₂ = 5.0g H₂ x 1 mole.

$$H_2/2.016_g H_2 = 2.48 \text{ mol } H_2$$

If you are given density and volume, first calculate mass of the gas; mass (g) = ρ(g/L) x V(L).

HOW TO USE THE IDEAL GAS LAW

Sample: Calculate the pressure for 2.5 moles of Ar gas occupying 3.5 liters at 25°C.

Make required changes to variables:

$$T(K) = 25^\circ C + 273.15 = 298.15K$$

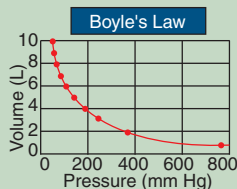
Calculate: P = nRT/V = 2.5 moles

$$\times 0.082L \text{ atm}/(\text{mol } K) \times 298.15K/3.5L$$

$$= 17.5 \text{ atm (Other units cancel)}$$

HOW TO USE BOYLE'S LAW

P & V are inversely proportional with constant T & n.



This is an **inverse-proportionality** problem:

$$P \propto 1/V$$

Given: V changes by a factor of z.

Calculate: P_{fin} = 1/z x P_{init}.

$$V \propto 1/P$$

Given: P changes by a factor of z.

Calculate: V_{fin} = 1/z x V_{init}

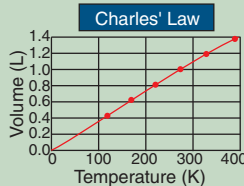
Sample: The pressure of a 4.0L sample changes from 2.5 atm to 5.0 atm. What is the final V?

Given: P changes 2-fold: 2.5 to 5.0 atm

Calculate: V_{fin} = 1/2 x V_{init} = 1/2 x 4.0L = 2.0L

HOW TO USE CHARLES' LAW

V and T are linearly proportional with constant n & P



This is a **direct-proportionality** problem: V ∝ T:

Given: T changes by a factor of z.

Calculate: V_{fin} = z x V_{init}.

Pitfall: T must be in Kelvin; convert °C to K.

Sample: A 3.5L sample of He gas is at 300 K; the T is raised to 900 K; what is the new V?

Given: T increases 3-fold: 300 to 900 K

Calculate: V_{fin} = 3 x V_{init} = 3.0 x 3.5 L = 10.5 L

WORKING WITH GASES cont.

HOW TO USE AVOGADRO'S LAW

V is proportional to n with constant T & P.

This is a **direct-proportionality** problem: $V \propto n$

Given: The # of moles changes by a factor of z.

Calculate: V_{fin} = z x V_{init}.

Sample: A 2.0 mole gas sample occupies 30.0L. Determine V for 1.0 mole of the gas.

Given: n changes by a factor of 1/2

Calculate: V_{fin} = 1/2 x 30.0 L = 15.0 L

HOW TO CALCULATE THE SPEED OF A GAS MOLECULE

Note: R = 8.314510 J/(K mol)

$$= 8.314510 \text{ kg m}^2/(\text{s}^2 \text{ K mol})$$

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

Sample: Calculate the vrms for He at 300 K

Given: T = 300;

He, M = 4.00 g/mol = 4.00 x 10⁻³ kg/mol

Calculate: v_{rms} = √(3 RT/ M)

$$= \sqrt{(3 \times 8.31451 \text{ kg m}^2/\text{s}^2/(\text{K mol})$$

$$\times 300 \text{ K}/4.00 \times 10^{-3} \text{ kg/mol})$$

$$= \sqrt{(1.87 \times 10^6 \text{ m}^2/\text{s}^2)} = 1,370 \text{ m/s}$$

Pitfall: Watch the units on R and M; the final unit is m/s. T must be in Kelvin.

HOW TO USE GRAHAM'S LAW OF EFFUSION

The **relative** rate of effusion for molecules of mass M₁ and M₂

$$\frac{\text{Rate } M_1}{\text{Rate } M_2} = \sqrt{\frac{M_2}{M_1}}$$

Sample: Determine relative

rate of effusion for H₂ and CO₂.

Given: M₁=M_{H2} = 2 g/mol; M₂=M_{CO2} = 44 g/mol

Calculate: Rate H₂/Rate CO₂ = √(44/2) = 4.7

H₂ diffuses 4.7 times as fast as CO₂

Pitfall: It is easy to invert the M₁/M₂; the smaller atom is always faster.

SOLIDS & LIQUIDS

CALCULATING MOLES OF REAGENTS

Determine the # of **moles** in "x-grams," or the **mass** needed to give a certain # of moles.

Required data: The **molar mass**.

Sample: Calculate # of moles in 5.6 g of NaCl.

Given: NaCl molar mass = 58.44 g

Calculate: Moles of NaCl

$$= 5.6g/58.44g/\text{mole} = 0.096 \text{ moles NaCl}$$

Sample: Calculate mass of 0.25 moles of NaCl.

Calculate: Mass of NaCl

$$= 0.25 \text{ moles} \times 58.44g/\text{mole}$$

$$= 14.61g \text{ NaCl}$$

HOW TO CALCULATE MASS OF LIQUIDS

Density (ρ), has units of g/mL.

Pure reagent: Use ρ & volume to determine the mass.

$$\text{vol} \times \rho = \text{mass}$$

Sample: Determine the mass of 30.0 mL of methanol.

Given: ρ = 0.790 g/mL

Calculate: Mass = vol x ρ = 30.0mL x 0.790 g/mL = 23.7 g

SOLIDS & LIQUIDS cont.

HOW TO CALCULATE MOLES OF REAGENTS IN SOLUTION

Common units: **Molarity (M)**: Moles of solute **per liter of solution**; **molarity (m)**, moles of solute **per kg of solvent**. Multiply the solution volume by the molarity to calculate **moles** of reagent.

Given: Solution molarity (M)

Calculate: Moles = vol x M

Pitfall: Volume should be in L. If you want to use "mL," denote M as "mmol/mL of solution"; **mmol = 0.001 mole**.

Sample: Determine the moles and mass of NaCl in 25 mL of 2.35 M NaCl solution.

Given: NaCl molarity = 2.35 M; molar mass

$$= 58.44 \text{ g}$$

Calculate: Volume = 25 mL x 1 L/1000 mL = 0.025 L

Calculate: NaCl moles = 2.35 mol/L x 0.025 L = 0.059 moles

Calculate: NaCl mass = 0.059 moles x 58.44 g/mole = 3.45 g

HOW TO PREPARE SOLUTIONS

A **solution** is prepared by dissolving a known mass of solid in a specific amount of solvent.

Sample: Prepare one liter of 1.00 M NaCl.

Given: NaCl molar mass = 58.44 g/mole

Step 1. Weigh out 58.44 g of NaCl and transfer to a 1 L volumetric flask.

Step 2. Dissolve the salt; fill with water to the 1-L line.

Note: If you need a different M, change the mass of NaCl.

PREPARING A DILUTE SOLUTION FROM A STOCK SOLUTION

Key: Conserve mass and moles. The molarity and volume of the stock and diluted solutions are governed by:

$$M_{\text{stock}} \times V_{\text{stock}} = M_{\text{dilute}} \times V_{\text{dilute}}$$

Sample: Prepare 50 mL of a 1.0 M solution from "a" mL of 2.0 M stock.

Given: M_{stock} = 2.0 M; V_{stock} = a;

$$M_{\text{dilute}} = 1.0 \text{ M}; V_{\text{dilute}} = 50 \text{ mL}$$

Calculate: "a" mL = 50mL x 1.0M/2.0M = 25mL

HOW TO CALCULATE COLLIGATIVE PROPERTIES

One example is **freezing point depression**:

$$\Delta T_f = -m \times K_f \times \text{ion factor}$$

m: Molality; K_f: solvent constant;

Ion factor: # of ions produced by solute: 1 for Molecular solute; 2 or more for ionic salt

Sample: Calculate the freezing point depression for a solution of 100g of NaCl in 500g of water.

Given: K_f (water) = 1.86 °C/m;

molar mass of salt = 58.44 g/mol

Ion factor = 2

Calculate: Mass of solvent = 500g x 1kg/1,000g = 0.500 kg

Calculate: m of NaCl

$$= (100.0g/58.44g/\text{mol})/0.500\text{kg} = 3.42 \text{ m}$$

Calculate: ΔT = - 3.42 m x 1.86 °C/m x 2

$$= -12.72 \text{ }^\circ\text{C}$$

THERMODYNAMICS: HEAT, DISORDER & EQUILIBRIUM

• ΔH (enthalpy), DG (free energy) and DS (entropy) characterize a process.

$$\Delta G = \Delta H - T \Delta S \text{ and } \Delta G = -RT \ln K_{eq}$$

HOW TO CALCULATE ΔG, ΔH, ΔS FROM STANDARD DATA

Given: ΔG_f⁰ (Free Energy of Formation), in kJ/mole; ΔH_f⁰ (Enthalpy of Formation), in kJ/mole;

S⁰ (Standard Entropy), J/(mole K)

Calculate: ΔG = sum of product ΔG_f⁰ - sum of reactant ΔG_f⁰

THERMODYNAMICS cont.

Calculate: ΔH = sum of product ΔH_f°
- sum of reactant ΔH_f°
Calculate: ΔS = sum of product S°
- sum of reactant S°

Sample: Calculate DH for the reaction:
 $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \Rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$.
Given: ΔH_f° -74.6 2 x 0.0 -393.5 2 x -285.8.
Calculate: ΔH = product ΔH_f° - reactant ΔH_f° .
 $\Delta H = -393.5 - 571.6 + 74.6 = -890.5$ kJ/mole.

Note: Combustion is an exothermic reaction.

Sample: Calculate ΔS for the phase change:
 $\text{H}_2\text{O}(\text{l}) \Rightarrow \text{H}_2\text{O}(\text{g})$
Given: S° 70.0 188.8
Calculate: $\Delta S = 188.8 - 70.0 = 118.8$ J/(mole K)
Note: A gas has more entropy than a liquid.

1. Does the reaction release or absorb heat? Examine DH.

- Exothermic (releases heat): $\Delta H < 0$
- Endothermic (absorbs heat): $\Delta H > 0$

2. Does the reaction proceed to completion? Is the reaction spontaneous? Examine DG.

- $\Delta G > 0$ not spontaneous $K_{eq} < 1$
- $\Delta G < 0$ spontaneous $K_{eq} > 1$
- Use ΔG to calculate K_{eq}

HOW TO CALCULATE K_{eq}

Sample: Calculate K_{eq} if the ΔG of a reaction is -10 kJ/mole at 25°C.
Given: $\Delta G = -10$ kJ; $T = 25^\circ\text{C}$; $R = 8.3145$ J/(K mole)
Calculate: $T(\text{K}) = 25^\circ\text{C} + 273.15 = 298.15$ K
Calculate: $\Delta G = -10$ kJ x 1,000 J/kJ = -10,000 J
Calculate: $K_{eq} = \exp(-(-10,000 \text{ J/mole}) / (8.3145 \text{ J/(K mole)} \times 298.15 \text{ K}))$
 $= e^{4.03} = 56.5$

The equilibrium shifts to the right, a **spontaneous reaction**.

Pitfall: T must be in K; make sure you are consistent with J and kJ.

HOW TO HANDLE THE "ADDITION" OF REACTIONS

1. Hess' Law: If you "sum" reactions, you also sum ΔH , ΔG and ΔS .

Sample: Calculate ΔH for the reaction $\text{A} + \text{D} \Rightarrow \text{F}$.
Given: $\text{A} + \text{B} \Rightarrow \text{C}$ $\Delta H = 50$ kJ/mole
Given: $\text{C} + \text{D} \Rightarrow \text{B} + \text{F}$ $\Delta H = 43$ kJ/mole
Sum of the reactions gives $\text{A} + \text{D} \Rightarrow \text{F}$
Calculate: $\Delta H = 50 \text{ kJ/mole} + 43 \text{ kJ/mole} = 93 \text{ kJ/mole}$

2. What happens if you reverse a reaction?

If you reverse the reaction, **change the sign** of ΔH , ΔG or ΔS .

Sample: Determine DH for the reaction $\text{C} \Rightarrow \text{A} + \text{B}$.
Given: $\text{A} + \text{B} \Rightarrow \text{C}$ $\Delta H = 50$ kJ/mole
For $\text{C} \Rightarrow \text{A} + \text{B}$, the reverse of this reaction
Calculate: $\Delta H = -50$ kJ/mole

3. How do equation coefficients impact the DH, DG or DS?

Thermodynamic properties scale with the coefficients.

Sample: Determine DH for the reaction:
 $2\text{A} + 2\text{C} \Rightarrow 2\text{D}$.
Given: $\text{A} + \text{C} \Rightarrow \text{D}$ $\Delta H = -50$ kJ
For $2\text{A} + 2\text{C} \Rightarrow 2\text{D}$, "double" this reaction.
Calculate: $\Delta H = 2 \times -50 \text{ kJ} = -100 \text{ kJ}$.

ACID-BASE CHEMISTRY

For simplicity: $[\text{H}_3\text{O}^+] = [\text{H}^+]$; $[\]$ refers to M, molarity; HAc = acetic acid; Ac^- = acetate

WHY WATER DISSOCIATES $\text{H}_2\text{O} \Leftrightarrow \text{OH}^- + \text{H}^+$

- $K_w = [\text{OH}^-][\text{H}^+] = 1 \times 10^{-14}$ at 25 °C
- For pure water: $[\text{OH}^-] = [\text{H}^+] = 1 \times 10^{-7}$ M
- Acidic solution: $[\text{H}^+] > 1 \times 10^{-7}$ M
- Basic solution: $[\text{H}^+] < 1 \times 10^{-7}$ M

HOW TO CALCULATE pH & $[\text{H}^+]$

$$\text{pH} = -\log_{10} [\text{H}^+] \quad [\text{H}^+] = 10^{-\text{pH}}$$

Sample: Determine the pH for a specific $[\text{H}^+]$.
Given: $[\text{H}^+] = 1.4 \times 10^{-5}$ M
Calculate: $\text{pH} = -\log_{10} [1.4 \times 10^{-5}] = 4.85$
 Sample: Determine $[\text{H}^+]$ from pH.
Given: $\text{pH} = 8.5$
Calculate: $[\text{H}^+] = 10^{-(8.5)} = 3.2 \times 10^{-9}$ M

HOW TO CALCULATE pOH & $[\text{OH}^-]$

$$\text{pOH} = -\log_{10} [\text{OH}^-] \quad [\text{OH}^-] = 10^{-\text{pOH}}$$

$$\text{pOH} + \text{pH} = 14 \text{ for any given solution}$$

Sample: Determine the $[\text{OH}^-]$ from pOH or pH.
Given: $\text{pH} = 4.5$
Calculate: $\text{pOH} = 14 - 4.5 = 9.5$
Calculate: $[\text{OH}^-] = 10^{-(9.5)} = 3.2 \times 10^{-10}$ M

HOW TO WORK WITH K_a & K_b

1. Acid $\text{HA} \Leftrightarrow \text{H}^+ + \text{A}^-$

- $K_a = [\text{H}^+]_{eq} [\text{A}^-]_{eq} / [\text{HA}]_{eq}$
- A^- : Conjugate base; $K_b(\text{A}^-) = K_w / K_a(\text{HA})$
- $\text{p}K_a = -\log_{10}(K_a)$

2. Base $\text{B} + \text{H}_2\text{O} \Leftrightarrow \text{BH}^+ + \text{OH}^-$

- $K_b = [\text{OH}^-]_{eq} [\text{BH}^+]_{eq} / [\text{B}]_{eq}$
- BH^+ : Conjugate acid; $K_a(\text{BH}^+) = K_w / K_b(\text{B})$
- $\text{p}K_b = -\log_{10}(K_b)$; weak bases have large $\text{p}K_b$

CALCULATING K_a

Substitute the experimental equilibrium concentrations into the K_a expression.

$$K_a = \frac{[\text{H}^+]_{eq} [\text{A}^-]_{eq}}{[\text{HA}]_{eq}}$$

Sample: Determine K_a and $\text{p}K_a$ from equilibrium concentration data for HA.
Given: $[\text{H}^+]_{eq} = 1 \times 10^{-4}$ M; $[\text{A}^-]_{eq} = 1 \times 10^{-4}$ M;
 $[\text{HA}]_{eq} = 1.0$ M
Calculate: $K_a = 1 \times 10^{-4} \times 1 \times 10^{-4} / 1.0 = 1 \times 10^{-8}$
Calculate: $\text{p}K_a = -\log_{10}(1 \times 10^{-8}) = 8.0$

HOW TO CALCULATE K_b & $\text{p}K_b$ OF CONJUGATE BASE (A^-) OF AN ACID HA

$$K_b(\text{A}^-) = K_w / K_a(\text{HA})$$

Sample: Determine K_b and $\text{p}K_b$ for the acetate ion, Ac^- .

Identify the acid: Acetic acid

Given: $K_a(\text{HAc}) = 1.7 \times 10^{-5}$
Calculate: $K_b(\text{Ac}^-) = K_w / K_a(\text{HAc})$
 $= 1 \times 10^{-14} / 1.7 \times 10^{-5} = 5.9 \times 10^{-10}$
Calculate: $\text{p}K_b(\text{Ac}^-) = -\log_{10}(5.9 \times 10^{-10}) = 9.23$

HOW TO CALCULATE % DISSOCIATION OF AN ACID

$$\% \text{ dissociation} = 100\% \times \frac{[\text{H}^+]_{eq}}{[\text{HA}]_{initial}}$$

Sample: Determine the % diss. for a 0.50 M HA that produces $[\text{H}^+]_{eq} = 0.10$ M and $[\text{HA}]_{eq} = 0.4$.
Given: $[\text{H}^+]_{eq} = 0.10$ M; $[\text{HA}]_{initial} = 0.50$ M
Calculate: $\% \text{ diss.} = 100\% \times 0.10 / 0.50 = 20\%$

Pitfall: Be sure to use $[\text{HA}]_{initial}$, not $[\text{HA}]_{eq}$.

ACID-BASE CHEMISTRY cont.

HOW TO CALCULATE $[\text{H}^+]_{eq}$

	HA	$\Leftrightarrow$	$\text{H}^+ + \text{A}^-$
Init	$[\text{HA}]_{init}$		0 0
Equil	$[\text{HA}]_{init} - a$		a a

Method: Substitute the "Equil" expressions into K_a and solve the quadratic equation:

$$\text{Start with: } K_a = a^2 / ([\text{HA}]_{init} - a)$$

$$\text{This rearranges to: } K_a \times ([\text{HA}]_{init} - a) = a^2$$

$$a^2 + K_a \times a - K_a \times [\text{HA}]_{init} = 0$$

Given K_a and $[\text{HA}]_{init}$, use the quadratic formula to obtain "a," $[\text{H}^+]_{eq}$.

Sample: Calculate $[\text{H}^+]$ for 0.5 M HAc.
Given: $K_a = 1.7 \times 10^{-5}$

Substitute into quadratic equation:

$$a^2 + K_a \times a - K_a \times [\text{HAc}]_{init} = 0$$

$$a^2 + 1.7 \times 10^{-5} \times a - 1.7 \times 10^{-5} \times 0.50 = 0$$

$$a^2 + 1.7 \times 10^{-5} \times a - 8.5 \times 10^{-6} = 0$$

Solve the quadratic equation:

$$a = [\text{H}^+]_{eq} = 0.0029 \text{ M}$$

Check your work: $K_a = (0.0029 \times 0.0029) / (0.5 - 0.0029) = 1.7 \times 10^{-5}$

Pitfall: Watch out for round-off error when you solve for the roots of the quadratic equation.

HOW TO CALCULATE THE $[\text{OH}^-]_{eq}$

	B + H ₂ O	$\Leftrightarrow$	$\text{BH}^+ + \text{OH}^-$
Init	$[\text{B}]_{init}$		0 0
Equil	$[\text{B}]_{init} - a$		a a

Method: Substitute the "Equil" expressions into K_b and solve the quadratic equation:

$$\text{Start with: } K_b = a^2 / ([\text{B}]_{init} - a)$$

$$\text{Solve this quadratic equation for "a" = } [\text{OH}^-]_{eq}$$

WHY DO SALTS HYDROLYZE?

Basic Salts react with water to form OH^- .

Sample: Sodium acetate: $\text{Ac}^- + \text{H}_2\text{O} \Leftrightarrow \text{HAc} + \text{OH}^-$

Acidic Salts react with water to form H_3O^+ .

Sample: Ammonium chloride:
 $\text{NH}_4^+ + \text{H}_2\text{O} \Leftrightarrow \text{NH}_3 + \text{H}_3\text{O}^+$

Neutral Salts do not react with water.

Sample: NaCl (product of strong acid + strong base)

HOW TO CALCULATE THE $[\text{H}^+]$ OR $[\text{OH}^-]$ FOR A SALT

Step 1: Is the salt acidic, basic or neutral?

Step 2: If acidic: Identify the weak acid, and the K_a ; if basic: Identify the weak base, and the K_b . If neutral: The solution will not have acidic or basic character.

Step 3: Set up the problem as a weak-base or weak-acid dissociation problem.

Given the initial salt concentration, calculate the equilibrium $[\text{H}^+]$ or $[\text{OH}^-]$.

Sample: Determine the $[\text{H}^+]$ or $[\text{OH}^-]$ for a 0.40 M NaAc solution.

Step 1: NaAc is a **basic salt**

Step 2: Ac^- is the base; $K_b(\text{Ac}^-) = K_w / K_a(\text{HAc}) = 5.9 \times 10^{-10}$

Step 3: Solve as a "weak-base dissociation" problem
 $[\text{B}]_{init} = 0.40$ M Ac^- ; calculate $[\text{OH}^-]_{eq}$

Pitfall: You must correctly identify the acid or base formed by the salt ions, and determine the K_a or K_b .

ACID-BASE CHEMISTRY cont.

HOW TO CALCULATE pH OF A BUFFER

Buffer of Weak Acid and Conjugate Base

Start with both weak acid, $[HA]_{init}$, and salt, $[A^-]_{init}$.

The equilibrium concentrations are governed by

$$K_a = [H^+]_{eq}[A^-]_{eq}/[HA]_{eq}$$

	HA	$\rightleftharpoons$	$H^+ + A^-$
Init	$[HA]_{init}$	0	$[A^-]_{init}$
Equil	$[HA]_{init}-a$	a	$[A^-]_{init}+a$

Method: Substitute the "Equil" expressions into K_a and solve the quadratic equation:

$$K_a = a \times ([A^-]_{init} + a) / ([HA]_{init} - a)$$

$$K_a \times [HA]_{init} - a \times K_a = a \times [A^-]_{init} + a^2$$

$$a^2 + a \times ([A^-]_{init} - K_a) - K_a \times [HA]_{init} = 0$$

Given: K_a , $[A^-]_{init}$ and $[HA]_{init}$, solve for the roots of the quadratic, $a = [H^+]_{eq}$

Sample: Determine the pH of a buffer of 0.5 M HAc and 0.3 M Ac⁻.

$$\text{Given: } K_a(\text{HAc}) = 1.7 \times 10^{-5}$$

$$\text{Quadratic: } 0 = a^2 + a \times (1.7 \times 10^{-5} + 0.3)$$

$$-1.7 \times 10^{-5} \times 0.5$$

$$0 = a^2 + a \times (0.3) - 1.7 \times 10^{-5} \times 0.5$$

$$\text{Solve quadratic: } a = [H^+] = 2.8 \times 10^{-5} \text{ M}$$

$$\text{Calculate: } \text{pH} = -\log_{10}(2.8 \times 10^{-5}) = 4.55$$

HOW TO USE THE HENDERSON-HASSELLBACH APPROXIMATION FOR BUFFER PH

Assume that "a" in the previous problem is $\ll [HAc]_{init}$ and $[Ac^-]_{init}$

$$\text{Henderson Hassellbach: } \text{pH} = \text{p}K_a + \log_{10} \left\{ \frac{[A^-]}{[HA]} \right\}$$

Sample: Examine previous buffer problem:

$$\text{Given: } [A^-] = [Ac^-] = 0.3 \text{ M; } [HA] = [HAc] = 0.5 \text{ M}$$

$$\text{p}K_a = \text{p}K_a(\text{HAc}) = 4.77$$

$$\text{Calculate: } \text{pH} = 4.77 + \log_{10}(0.3/0.5)$$

$$= 4.77 - 0.22 = 4.55$$

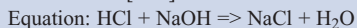
The approximation works.

HOW TO DO AN ACID-BASE TITRATION

A systematic acid-base neutralization used to determine the concentration of an unknown acid or base. At the equivalence point, moles of acid = moles of base.

Sample: The titration of 50.00 mL of an HCl solution requires 25.00 mL of 1.00 M NaOH.

Calculate the [HCl].



This gives 1:1 molar ratio of HCl: NaOH.

At the **equivalence point**: The moles balance, or more conveniently: Mmoles HCl = mmoles NaOH.

$$M(\text{HCl}) \times \text{vol-acid (mL)} = \text{vol-base (mL)} \times M(\text{NaOH})$$

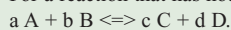
$$M(\text{HCl}) = \text{vol-base (mL)} \times M(\text{NaOH}) / \text{vol-acid (mL)}$$

$$\text{Calculate: } M(\text{HCl}) = 25.00 \text{ mL} \times 1.00 \text{ M} / 50.00 \text{ mL} = 0.50 \text{ M HCl}$$

Pitfall: Watch the units on volume and molarity work with "L & mole" or "mL & mmole."

EXAMINATION OF CHEMICAL EQUILIBRIUM

For a reaction that has not gone to completion:



At **equilibrium**, the process is described by the **equilibrium constant**, K_c .

$$K_c = \frac{[C]_{eq}^c [D]_{eq}^d}{[A]_{eq}^a [B]_{eq}^b}$$

For all other conditions, the process is described by the **reaction quotient**, Q_c :

$$Q_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

if $Q_c = K_c$, the reaction is at equilibrium

if $Q_c > K_c$, the reaction will go to the left

if $Q_c < K_c$, the reaction will go to the right

Gas-phase reactions may be described with K_p , based on reagent partial pressures. These calculations follow the same strategy as K_c .

EXAMINATION OF CHEMICAL EQUILIBRIUM cont.

HOW TO DETERMINE IF THE REACTION IS AT EQUILIBRIUM

Compare Q_c with K_c

Sample: For the reaction: $A \rightleftharpoons C$, $K_c = 0.60$; the observed $[A] = 0.1$ and $[C] = 0.20$.

Is the reaction at equilibrium? If not, predict the shift.

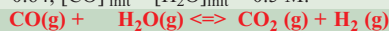
$$1. Q_c = [C]/[A] = 0.20/0.10 = 2.0; K_c = 0.60.$$

2. $Q_c > K_c$; process is not at equilibrium, it will shift to the left.

HOW TO PREDICT EQUILIBRIUM CONCENTRATIONS

Sample: Calculate the equilibrium concentrations for the following gas-phase reaction data:

$$K_c = 0.64; [CO]_{init} = [H_2O]_{init} = 0.5 \text{ M:}$$



	CO(g)	H ₂ O(g)	$\rightleftharpoons$	CO ₂ (g)	H ₂ (g)
Init	$[CO]_{init}$	$[H_2O]_{init}$	0	0	0
Equil	$[CO]_{init}-a$	$[H_2O]_{init}-a$	a	a	a

Note: The change "a" is the same for each because of the 1:1:1:1 coefficients in the equation.

Identify equilibrium expression:

$$K_c = [CO_2]_{eq}[H_2]_{eq}/[CO]_{eq}[H_2O]_{eq}$$

Substitute "equil" values:

$$K_c = a^2 / \{([CO]_{init} - a) \times ([H_2O]_{init} - a)\}$$

$$0.64 = a^2 / \{(0.50 - a)(0.50 - a)\}$$

Take square root of each side:

$$0.8 = a / (0.5 - a) \text{ or } -0.8 = a / (0.5 - a)$$

$$a = 0.222 \text{ or } a = -1.2$$

Use the first option, since "a" must be positive.

$$[CO_2]_{eq} = [H_2]_{eq} = 0.222 \text{ M}$$

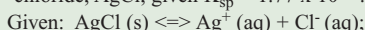
$$[CO]_{eq} = [H_2O]_{eq} = 0.50 - 0.222 = 0.278 \text{ M}$$

Check your work: $K_c = (0.222^2) / (0.278^2) = 0.64$

Pitfall: Watch out for round-off error; take the root that gives positive concentrations.

HOW TO DETERMINE THE SOLUBILITY LIMIT FROM K_{sp} (GIVEN IN MOLES/L OR G/L)

Sample: Determine the solubility limit for silver chloride, AgCl, given $K_{sp} = 1.77 \times 10^{-10}$.



$$K_{sp} = [Ag^+][Cl^-] = 1.77 \times 10^{-10}$$

Given: AgCl molar mass = 143.32 g/mole

At equilibrium, $[Ag^+]_{eq} = [Cl^-]_{eq} = \sqrt{K_{sp}}$

$$\text{Calculate: } [Ag^+]_{eq} = \sqrt{(1.77 \times 10^{-10})}$$

$$= 1.33 \times 10^{-5} \text{ M AgCl}$$

This is also $[AgCl]_{eq}$, the **molar solubility limit** for AgCl.

Calculate: The AgCl **g/L solubility limit**

$$= [AgCl]_{eq} \times \text{molar mass of AgCl}$$

$$= 1.33 \times 10^{-5} \text{ moles/L AgCl} \times 143.32 \text{ g/mole}$$

$$= 1.9 \times 10^{-3} \text{ g/L}$$

KINETICS & MECHANISMS

The goal of a kinetic study is to measure the **reaction rate**, determine the **rate constant**, k , the **rate law** and the **activation energy** (E_a).

HOW TO DETERMINE THE REACTION RATE

For " $A \Rightarrow B$," the **reaction rate** is the rate of appearance of product, $\Delta[B]/\Delta\text{time}$; or, rate of loss of reactant: $-\Delta[A]/\Delta\text{time}$.

Sample: How would you characterize the rate of: $CaCO_3(s) \Rightarrow CaO(s) + CO_2(g)$?

Answer: Focus on rate of CO_2 production;

$$\text{Rate} = \Delta[CO_2]/\Delta\text{time}.$$

HOW TO DETERMINE THE RATE LAW

The **rate law** gives the **order** of the reaction based on the steps in the overall reaction " $A + B \Rightarrow C$."

$$1. \text{Rate} = k[A], \text{ for a first-order reaction.}$$

$$2. \text{Rate} = k[A]^2, \text{ or } k[A][B], \text{ for a second-order reaction.}$$

$$3. \text{Rate} = k[A]^0, \text{ for a zero-order reaction.}$$

Pitfall: Equation coefficients describe the balanced overall reaction, not the mechanism and rate law.

KINETICS cont.

Two common rate-law methods: "**Initial rates**" and "**integrated rate equations**." Consider the reaction " $A \Rightarrow B$," with a rate law of the form: $\text{Rate} = k[A]^x$. The goal of kinetic study: Determine " x ," the order of the reaction.

1. Initial Rate Strategy:

Step 1: For $[A]_1$, measure the time required to produce $\Delta[B]$ of product.

$$\text{Calculate: } \text{Rate}_1 = \Delta[B]/\text{time}$$

Step 2: Measure the new reaction rate, rate_2 , for a different concentration, $[A]_2$.

The interplay of rate, $[A]$ and x , are governed by the equation: $\frac{\text{Rate}_1}{\text{Rate}_2} = \left\{ \frac{[A]_1}{[A]_2} \right\}^x$

Sample: Determine " x " if doubling $[A]$ also doubles the rate:

$$1. \text{Rate}_1/\text{Rate}_2 = 2$$

$$2. [A]_1/[A]_2 = 2$$

$$3. 2 = 2^x, x = 1, \text{ this is a 1st order process}$$

Sample: Determine " x " if doubling $[A]$ increases the rate by 4-fold:

$$1. \text{Rate}_1/\text{Rate}_2 = 4$$

$$2. [A]_1/[A]_2 = 2$$

$$3. 4 = 2^x, x = 2, \text{ this is a 2nd order process}$$

2. Integrated Rate Equation Strategy

Analyze " $[A]$ vs. time" data for the reaction.

The reaction is 1st order if the " $\ln[A]$ vs. t " graph is linear.

The reaction is 2nd order if " $1/[A]$ vs t " graph is linear.

In each case, k is the slope of the line.

HOW TO DETERMINE THE ACTIVATION ENERGY, E_a

$$\ln(k_1/k_2) = -E_a/R(1/T_1 - 1/T_2)$$

$$= E_a/R(T_1 - T_2)/(T_1 \times T_2)$$

Applications:

a. Predict k_1 at T_1 , given k_2 at T_2 and E_a .

b. Determine E_a from k_1 , k_2 , T_1 , T_2 ; only have to worry about k_1/k_2 .

Sample: The rate constant doubles when the temperature changes from 25.0°C to 50.0°C. What is E_a ?

$$\text{Given: } k_1/k_2 = 2$$

$$\text{Calculate: } T_1 = 50.0^\circ\text{C} + 273.15 = 323.2 \text{ K}$$

$$\text{Calculate: } T_2 = 25.0^\circ\text{C} + 273.15 = 298.2 \text{ K}$$

$$\text{Calculate: } \Delta T = T_1 - T_2 = 25.0 \text{ K}$$

$$\text{Calculate: } E_a = R \ln(k_1/k_2) T_1 \times T_2 / \Delta T$$

$$= 8.314 \text{ J/mole K} \times \ln(2) \times 323.2 \text{ K} \times 298.2 \text{ K} / 25.0 \text{ K}$$

$$= 22,200 \text{ J/mole} = 22.2 \text{ kJ/mole}$$

Pitfall: T must be in Kelvin; if you use the equation with " $1/T_1 - 1/T_2$," beware of round-off error in calculating inverse T .

KINETICS & EQUILIBRIUM

An equilibrium is characterized by competing forward and reverse reactions. The forward and reverse **rate constants** (k_f and k_r) are related to the equilibrium constant, K_{eq} . At equilibrium: The forward and reverse reaction rates are **equal**.

$$k_{eq} = k_f / k_r$$

ORGANIC CHEMISTRY

FUNDAMENTALS

TYPES OF ORGANIC COMPOUNDS

HYDROCARBON	-O- ADDED	>C=O ADDED	>COO ADDED	NITROGEN ADDED	SULFUR ADDED
ALKANE $\text{C}-\text{H}$ - ethane: C_2H_6 - methyl (Me): $-\text{CH}_3$ - ethyl (Et): $-\text{C}_2\text{H}_5$ ALKENE $\text{C}=\text{C}$ - ethene: C_2H_4 - diene: two $\text{C}=\text{C}$ - triene: three $\text{C}=\text{C}$ ALKYNE $-\text{C}\equiv\text{C}-$ - ethyne: C_2H_2 AROMATIC  - benzene: C_6H_6 - arene: C_6H_5 (Ar-)	ALCOHOL $\text{R}-\text{OH}$ - methanol: Me-OH (methyl alcohol) - phenol: Ar-OH - diol/glycol: (2 -OH) - glycerol: (3 -OH) ETHER $\text{R}-\text{O}-\text{R}$ - ethoxyethane: Et-O-Et (diethyl ether) EPOXY  - cyclic ether PEROXIDE $\text{R}-\text{O}-\text{O}-\text{R}'$	ALDEHYDE  - methanal: H_2CO (formaldehyde) - benzaldehyde: Ar-CHO KETONE  2-propanone: Me-CO-Me (dimethyl ketone, acetone) - diketone: R-CO-R''-CO-R'	CARBOXYLIC ACID  - ethanoic acid: Me-COOH (acetic acid) - acetate ion: Me-COO⁻ - benzoic acid: Ar-COOH Dicarboxylic acid HOOC-R-COOH ESTER  - ethyl acetate: Me-CO-OEt, Me-CO-OEtH, Other derivatives: - Peroxyacid: R-CO-OOH - Acid anhydride: RCO-O-CO-R'	AMINE $\text{R}-\text{N}-\text{R}$ - methyl amine: $\text{H}_3\text{C}-\text{NH}_2$ - phenylamine: Ar-NH₂ (aniline) - R-NH₂ (1°), RR'NH (2°), RR''N (3°) NITRO R-NO ₂ DIAZO R-N≡N NITRILE $\text{R}-\text{C}\equiv\text{N}$ - methane nitrile: Me-CN AMIDE  - acetamide: Me-CO-NH₂	- thiol: R-SH - thioether: R-S-R' - disulfide: R-S-S-R' - thiol ester: R-CO-SR' - sulfoxide: R-SO-R' - sulfone: R-SO₂-R' - sulfonic acid: R-SO₃H HALOGEN ADDED - haloalkane: R-X: Me-Cl chloromethane - halobenzene: Ar-X chlorobenzene: Ar-Cl - acyl halide: R-CO-X - aryl halide: Ar-X

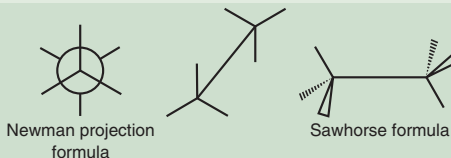
FORMULAS AND ISOMERS

Molecular formula: elemental symbols with subscripts denote the composition of a compound

Empirical formula: subscripts denote the relative elemental composition

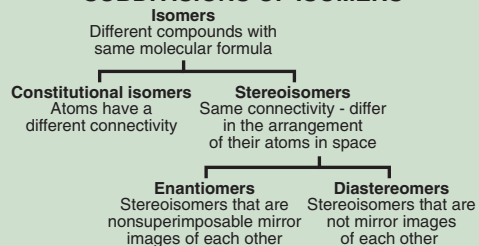
Graphical depiction:

- Dash formula: diagram all atoms, bonds as dashes
- Bond line formula: hide H, show carbon skeleton as lines, other atoms explicit



- Newman Projection: 2-d depiction
- 3-dimensional: wedges of sawhorse denote structure

SUBDIVISIONS OF ISOMERS



constitutional isomers: different bonding connectivity (ex. rings, bonds, branching, substituent positions)

tautomers: easily interconverted structural isomers (ex. keto-enol for ketone)

chiral: not identical with mirror image

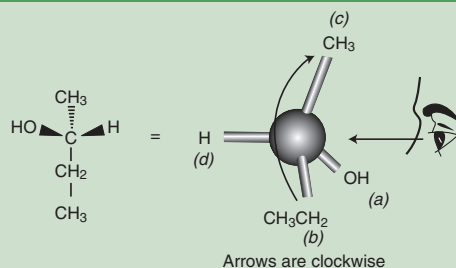
achiral: has plane of symmetry (superimposable on mirror-image)

epimers: a pair of diastereomers which differ only in the configuration of one atom

More than 1 chiral center:

- n chiral centers, $\leq 2^n$ stereoisomers
- meso:** two chiral centers, 4 isomers: 3 stereoisomers, 1 achiral (mirror-plane)

FORMULAS AND ISOMERS

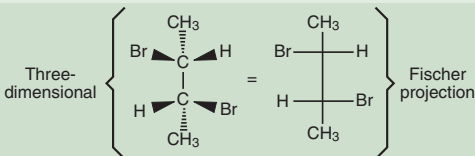


R/S notation: the four different atoms or groups attached to a central atom are ranked a,b,c,d, by molar mass. The lowest (d) is directed away from the viewer and the sequence of a-b-c produces clockwise (R) or counter-clockwise (S) configuration.

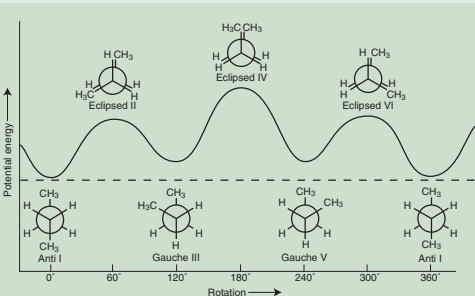
- chiral (optically active):** + or - rotation of plane polarized light. R/S: opposite effects
- racemic:** 50/50 mixture of stereoisomers (no net optical activity)
- nomenclature:** note R/S and +/- in the compound name; example: R (+) bromochloromethanol.

Fisher-projection: diagram depicts chiral/3-D structure

molecular conformations: molecule exhibits structural variation due to free rotation about C-C single bond



Newman-diagram: depict rotation about a C-C bond; **eclipsed** (high energy), **anti** (low energy), **gauche** (intermediate energy)



COMMON TERMS

aliphatic: non-aromatic

aromatic: benzene ring

heterocyclic: non-carbon atom in the ring structure

hydrocarbon: compound of H and C

paraffin: alkane

olefin: alkene

saturated: maximum # of H's (all C-C single bonds)

unsaturated: at least one C-C multiple bond

NOMENCLATURE

IUPAC - standard guidelines for naming compounds

Nomenclature Strategy - find longest carbon chain, identify and note location of functional groups and substituents by chain position number.

Classes of compounds are defined by the functional group. There are many common names and functional group names. Multiple names are possible.

CARBON CHAIN PREFIXES

# of C's	Prefix	R-group
1	meth-	methyl
2	eth-	ethyl
3	prop-	propyl
4	but-	butyl
5	pent-	pentyl
6	hex-	hexyl
7	hept-	heptyl
8	oct-	octyl
9	non-	nonyl
10	dec-	decyl

cyclo-: ring structure; example: cyclopropane 3-carbon ring molecule

iso-: two methyl groups on the terminus of a chain

n-: normal straight chain

t-: tertiary alkyl group

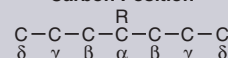
vic (vicinal): two substituents on adjacent carbons

gem (geminal): two substituents on the same carbon

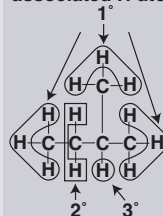
alkene isomers: cis or trans

benzene substitution positions: ortho(1,2), meta (1,3), para (1,4)

Carbon Position



Carbon atoms & associated H-atoms



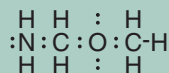
CHEMICAL BONDING IN ORGANIC COMPOUNDS

Bonds are usually **polar covalent**. Polarity arises from **electronegativity** difference; the larger the difference, the more polar the bond. The more electronegative atom is the negative end of the bond.

In $>C=O$, O is negative, C is positive.

LEWIS STRUCTURE: SIMPLEST MODEL

- Assign valence electrons as bonding electrons and non-bonding lone pairs.
- Octet rule**: each atom is assigned 8 electrons; except H (2) and atoms with d-orbitals (the "filled-shell rule")



Bond Order (BO): # of bonds divided by the # of bonded neighbors. For a given pair of atoms, increased bond order reflects a stronger, shorter bond.

Example:	BO	Length (Å)	Energy (Kcal/mole)
C-C	1	1.53	90
C=C	2	1.33	173
C≡C	3	1.20	230

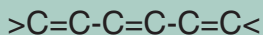
Formal charge (effective atomic charge):

- = (# of non-bonded electrons) + (1/2 # of bonded electrons) - (# of atomic valence electrons)
- The ideal formal charge of each atom is zero. Otherwise, minimize magnitude of charge by shifting charge to the more electronegative atom (especially for ions).

RESONANCE

The "average" of several Lewis structures provides a more accurate view of the bonding. Example: CO_3^{2-} has 3 equal bonds, though each of 3 Lewis structures has 1 double bond and 2 single bonds.

- delocalization**: resonance lowers the energy; electrons are dispersed, diminishing electron-electron repulsion
- conjugated alkene**: has alternate single/double bonds:



- Hückel Rule**: A planar cyclic molecule with $(4n+2)$ π -electrons is aromatic.

Ex: Benzene



- antiaromatic**: $4n$ π -electrons: aromatic excited state.
 - heterocyclic**: heteroatom lone pairs join the planar π system
- Examples:



Pyridine



Pyrrole



Furan

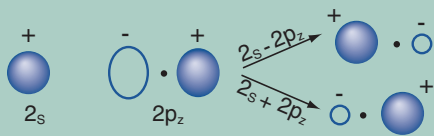


Thiophene

REFINED MODEL: VALENCE BOND THEORY

Overlap of atomic orbitals (AO's) or hybrids allows electrons to pair up, forming a chemical bond.

hybrid orbitals: valence AO's mix to accommodate "equivalent" bonded neighbors. Non-hybridized orbitals form lone pairs or π bonds.



MOLECULAR STRUCTURE AND HYBRID AO'S

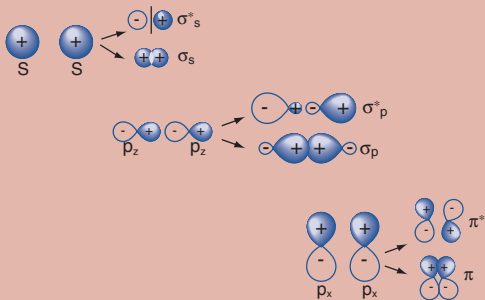
VSEPR (Valence Shell Electron Pair Repulsion): bonding pairs (X) and lone pairs (E) define geometry of AX_n ; reflects hybridization of A

- sp^3 - AX_4 : tetrahedral, bond angle of 109.4° ; alkane; lone-pair larger than bonded pair, distorts geometry
Ex: AX_3E pyramidal; amines, NR_3 , ammonia; AX_2E_2 bent: water: alcohol: R-O-H , ether: $\text{R-O-R}'$
- sp^2 - AX_3 trigonal planar (120°); C-C-C in aromatic ring;
Ex: R-CO-R in ketone, aldehyde, carboxylic acid
- sp - AX_2 linear;
Ex: alkyne $\text{-C}\equiv\text{C-}$; nitrile $\text{R-C}\equiv\text{N}$

QUANTUM MECHANICAL MODEL: MO THEORY

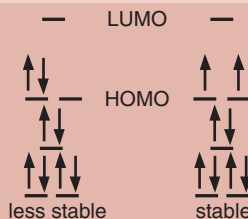
The **Schrodinger Equation**: $\hat{H}\psi = E\psi$, gives energy (E) and wavefunction (ψ); $\hat{H}$, Hamiltonian, the energy operator. ψ determines total energy, electron density and orbital energies.

- ψ is given by a set of MO's (**molecular orbitals**) formed by combining AO's. Each MO creates an energy level for two electrons.
- Constructive overlap of AO's : **bonding** = stable
- Destructive overlap of AO's : **anti-bonding** = unstable
- On-bond-axis: σ bonding ; σ^* antibonding
- Off-bond-axis: π bonding ; π^* antibonding
- Organic compound: focus on s and p orbitals
- Transition-metal atom: focus on d orbitals



MO'S AND ENERGY

- # of MO's = # of AO's
- bond order**: # of bonding electron-pairs - the # of antibonding electron-pairs
- Aufbau & Pauli Exclusion Principles**: MO's are filled from lower to higher energy; each level holds up to two electrons with paired spins
- HOMO**: Highest Occupied MO
- LUMO**: Lowest Unoccupied MO
- Hund's Rule**: For MO's of equal energy, maximize the total electron spin



APPLICATIONS OF MO THEORY

Chemical Reactivity: The electrons in the **HOMO** are most likely to participate in nucleophilic attack (electron donor). These are the least stable (most reactive) valence electrons in the molecule.

- The **LUMO** is likely to represent an electrophilic site (electron acceptor). In photochemical processes electrons are excited to the LUMO.
- Molecules with **unpaired** electrons in MO levels exhibit a net electron spin which can be measured by **electron spin resonance spectroscopy (ESR)**.

INTERMOLECULAR FORCES

- ELECTROSTATIC INTERACTIONS**: strong forces between ions; for charges q_1 and q_2 ; separated by r_{12} , and solvent dielectric constant, ϵ

$$\text{Energy} = \frac{1}{\epsilon} \frac{q_1 \cdot q_2}{r_{12}}$$

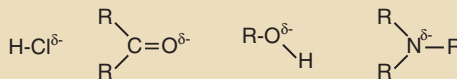
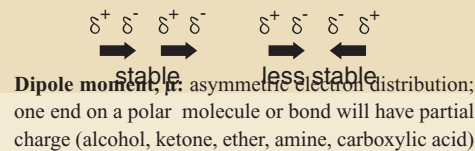
Solvent with large ϵ stabilizes carbocation, carbanion

Solvents (ϵ): water (80), ethanol (25), ethyl ether (4), methanol (33), acetone (21), hexane (1.9), benzene (2.2), toluene (2.4), phenol (9.8), aniline (6.9), pyridine (12), CCl_4 (2)

- LONDON FORCES** (dispersion): attraction due to induced dipole moments; increases with α

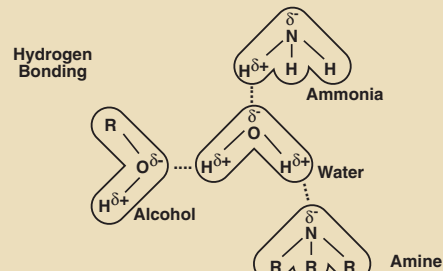
Polarizability, α : measures distortion of electron cloud by electric field of other nuclei and electrons

- DIPOLE-DIPOLE INTERACTION**: the positive end of one dipole is attracted to the negative end of another dipole. Increases with μ .



HYDROGEN BONDING

Enhanced dipole-dipole interaction between bonded H and the lone-pair of neighboring O, N or S. Can lead to dimer formation; gives "structure" to polar liquids.



hydrophobic ("water-fearing"): repelled by a polar group; attracted to "fat" or a nonpolar group

Examples: alkane or alkyl group, arene, alkene

hydrophilic ("water-loving"): attracted to a polar group; repelled by a nonpolar group

Examples: -OH of alcohol, -NH of amine, -COOH of carboxylic acid

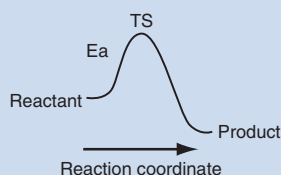
IMPACT ON SOLUBILITY

Ionic material tends to dissolve in **water**, as do polar organic compounds, R-OH , R-COOH , R-NH_2 .

- Non-polar** compounds are usually **insoluble** in water, but tend to dissolve in non-polar solvents: alkanes, alkenes, alkynes, aromatics.
- solvation**: process in which solute is surrounded by solvent molecules, creating a more stable system
- miscible** (2 or more substances form 1 phase): liquids with similar molecular properties (polar+polar, non-polar+non-polar)
- immiscible** (separate phases): aqueous and organic layers do not mix
- Compounds are partitioned between the layers based on chemical properties (acid/base, polar, nonpolar, ionic).
- partition coefficient**: the ratio of the solubility limits of a material in two immiscible phases

ORGANIC REACTIONS

- Most reactions take place in several simple steps, producing an overall **mechanism**.
- Incomplete reactions may establish equilibria.
 - Each step passes through an energy barrier, characterized by an unstable configuration termed the **transition state (TS)**.
 - The height of the barrier is the **activation energy (Ea)**.
- The slowest step in the mechanism, the **rate-determining step**, limits the overall reaction rate.
- Key principle:** examine the reactants and identify the points of excess and deficit electrons; organic reactions are best understood by "following the electrons."
- The electron movement is often described using an **arrow** in the reaction mechanism.



ACIDS AND BASES

Model Term	Acid	Base
Arrhenius	aqueous H_3O^+	aqueous OH^-
Bronsted-Lowry	proton donor	proton acceptor
Lewis	electron-pr. acceptor electrophiles	electron-pr. donor nucleophiles

Organic reactions: use Bronsted-Lowry and Lewis models

Acid $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$

- $K_a = [\text{A}^-][\text{H}^+]/[\text{HA}]$
- $\text{p}K_a = -\log_{10}(K_a)$
- strong acid:** full dissociation; examples HCl , H_2SO_4 and HNO_3
- weak acid:** $K_a \ll 1$, large $\text{p}K_a$; organic acid: RCOOH
- Examples ($\text{p}K_a$): acetic (4.75), carbonic (6.37), HF (3.45), HCN (9.31), benzoic (4.19), citric (3.14), formic (3.75), oxalic (1.23)
- Proton donor:** acetylene (25), ethanol (16), phenol (9.9)

Base $\text{BOH} \rightleftharpoons \text{B}^+ + \text{OH}^-$

- $K_b = [\text{OH}^-][\text{B}^+]/[\text{BOH}]$
- $\text{p}K_b = -\log_{10}(K_b)$
- strong base:** full dissociation; examples NaOH , KOH
- organic base:** R-NH_2
- weak base:** $K_b \ll 1$, large $\text{p}K_b$
- Examples: ($\text{p}K_b$): NH_3 (4.74), CN^- (4.7), hydrazine N_2H_4 (5.77), hydroxylamine (7.97), aniline (4.63), pyridine (5.25)

amphoteric: material which can react as an acid or a base. Example: **amino acid**; amine (base) and carboxylic acid functionality

zwitterion: self-ionization of the amino acid; the "acid" donates a proton to the "base"

OXIDATION-REDUCTION

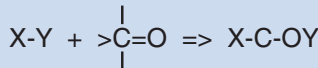
- oxidation:** loss of electrons; in organic reactions, add oxygen or remove hydrogen; examples: $\text{R} \Rightarrow \text{ROH} \Rightarrow \text{>C=O} \Rightarrow \text{RCOOH}$
- reduction:** gain of electrons; in organic reactions, add hydrogen or remove oxygen; examples: hydrogenation of alkene/alkyne to alkane

ORGANIC REACTIONS

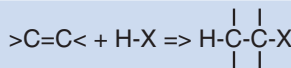
ADDITION REACTIONS

Add groups to a pair of atoms joined by a multiple bond; Ex: hydrogenation, halogenation, hydrohalogenation, hydration, hydroxylation. Two major types:

- nucleophilic:** nucleophile attacks C of >C=O



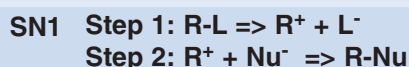
- electrophilic:** π electrons donated to electrophile; forms carbocation, which may rearrange



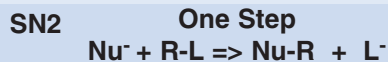
SUBSTITUTION REACTIONS

Replace existing group on an alkane or aromatic compound.

- Nucleophilic substitution:** nucleophile (Nu^-) seeks a "+" center (C of R group or >C=O), displaces leaving group -L. **SN1 and SN2 mechanisms**



SN1 : Favored for sterically hindered R; carbocation is stabilized by polar solvent ($3^\circ > 2^\circ > 1^\circ$), therefore carbocation may rearrange; racemic mixture; first-order kinetics (formation of R^+ determines the reaction rate).



SN2 : Backside attack of C bonded to L (the leaving group), inversion of stereochemical configuration; second order kinetics (Nu attack sets rate).

- Nucleophilic aromatic substitution:**

Two possible mechanisms:

- elimination/addition via benzyne intermediate (dehydrobenzene), Ex.: $\text{Ar-Cl} \Rightarrow \text{Ar-OH}$
- addition/elimination (SNAr) mechanism; electron-withdrawing groups facilitate nucleophilic attack; ex: nitrochlorobenzene $\Rightarrow$ nitrophenol

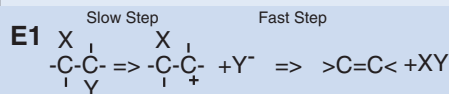
- Electrophilic aromatic substitution:**

- Electrophile, E^+ , attacks π electrons on the benzene ring, form arenium cation (ring stabilizes positive charge)
- H leaves, -E is attached to the ring
ex: alkylation, nitration, halogenation of benzene

ELIMINATION REACTIONS

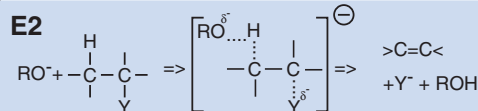
Reverse of addition, remove molecule "XY" from adjacent atoms, produces double bond.

Example: dehydrogenation, dehydrohalogenation, dehydration
Two possible mechanisms: E1 and E2



E1 : slow step: Y- leaves, forms a carbocation which may rearrange;

fast step: X leaves, giving alkene; 1st order kinetics



E2 : Concerted reaction; base partially bonds to -H, weakens bond to Y, Y departs and H is removed by the base, producing alkene; 2nd order kinetics

ORGANIC REACTIONS

HYDROLYSIS REACTIONS

A type of solvolysis where water (the solvent) breaks a bond; adds -H and -OH to the molecule (or -H and -OR when solvent is alcohol).

Example: **saponification:** base-hydrolysis of ester

CONDENSATION REACTIONS

Two reagents combine via bridging O or N, produce water or alcohol molecule;

Example: peptide bond ($\text{N-H} + \text{RCOOH}$), nylon synthesis, formation of polysaccharide

REARRANGEMENTS

Change in bond connectivity; common with radical, carbocation and carbanion intermediates.

- Driving force:** Bonds are altered to shift charge to a more substituted carbon; ex: resonance stabilization
- Carbocation stability: $\text{Ar} > 3^\circ > 2^\circ > 1^\circ$ carbons

IONIC REACTIONS

Heterolytic cleavage of $\text{X-Y} \Rightarrow \text{X}^+ + \text{Y}^-$; ion pair, stabilized by resonance or polar solvent. Characteristic of ionic reactions involving nucleophiles and electrophiles.

RADICAL REACTIONS

- Homolytic cleavage** of bond $\text{X-Y} \Rightarrow \text{X}^\bullet + \text{Y}^\bullet$
- radical:** Reactive species with unpaired electrons
- Reaction steps:** Initiation, propagation and termination. Radical geometries tend to be planar (sp^2 hybrid). Example: halogenation of alkane or alkene
- Radical stabilized by delocalization and rearrangement; relative stability: $\text{Ar-C}^\bullet\text{H}_2 > \text{R}_2\text{C}=\text{C}^\bullet\text{H}_2 > (\text{CH}_3)_3\text{C}^\bullet > (\text{CH}_3)_2\text{C}^\bullet\text{H} > \text{CH}_3\text{C}^\bullet\text{H}_2$

EXAMPLES OF SPECIFIC REACTIONS

acylation: add RCO-

alkylation: add -R Ex: **Grignard** (RMgX)

cyclization reaction:

Diels-Alder: diene + alkene/alkyne

decarboxylation: lose CO_2 from a carboxylic acid

hydroxylation: add -OH

nitration: add -NO_2

pyrolysis: anaerobic thermal decomposition

sulfonation: add $\text{-SO}_3\text{H}$

Wittig: $\text{>C=O} \Rightarrow \text{>CH}_2$

MECHANISM TERMS

anti addition: add to opposite faces of substrate

carbene: divalent carbon; ethylene radical: $\text{H}_2\text{C}^\bullet$

carbocation: trivalent carbon, positive formal charge

carbanion: negative formal charge on carbon

electrophile: a Lewis acid; attracted to the electron density found in a chemical bond or lone pair

endo: prefix for closed structure-type

exo: prefix for open structure-type

nucleophile: a Lewis base; attracted to the + charge of a nucleus or cation

oxonium: positively charged oxygen species

syn addition: add to the same face of a substrate

ylide: a neutral molecule with a formally-charged C⁻ next to a P⁺, or an electropositive heteroatom

SPECTROSCOPY AND INSTRUMENTAL METHODS

Chemical insight is gained by analyzing the interaction of matter and electromagnetic radiation (characterized by the wavelength, λ or frequency, ν).

MEASUREMENT METHODS

core electrons (X-ray) electronic transitions (UV/Vis)
vibrations (IR) nuclear spin (RF)

Energy of radiation is quantized in photons, $e = h\nu$; one photon excites one molecule to a higher energy state.

X-RAY

- Structure determination:** x-ray λ is comparable to atomic-spacing, scattered x-rays give a diffraction pattern characteristic of a crystal structure
- Photo-electron-spectroscopy (PES):** x-rays are energetic enough to dislodge core-electrons. Analysis of ejected electron energies gives MO and AO energies

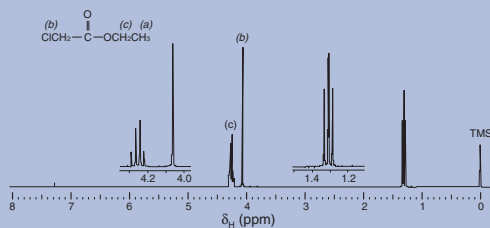
ULTRAVIOLET/VISIBLE

- Probes electronic transitions; peaks are broadened by rotational, vibrational and solvent effects. The size of the peak depends on electronic energy spacing.
- For organic molecules, often corresponds to a transition from a π -type HOMO to a π^* -type LUMO.

Colorimetry - Beer-Lambert Law: $A = abc$

Where **A** = absorbance; **a** = molar absorptivity (varies with λ); **b** = sample path length; **c** = molar concentration. **A** is related to transmission (**T**) by the equation: $A = -\log_{10}(T)$.

NUCLEAR MAGNETIC RESONANCE (NMR)



- RF radiation** (radio waves) matches the spacing between nuclear-spin energy levels **artificially** split by a strong magnetic field.
- The **resonance** is characteristic of an atom's chemical environment; given as δ , in ppm, the shift relative to a reference compound; for ^1H -NMR, TMS (tetramethylsilane).
- shielding:** resonance shifts to greater magnetic field (larger delta, δ) due to chemical environment of the atom. Proton NMR is most common, though isotopes of C, O, F, Si can be studied as well.

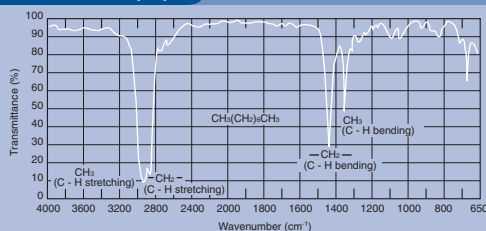
Shift ranges (in ppm)

R(^1H)	R($2^{\circ}, 3^{\circ}$)	R-X	ether	H-C $\equiv$ C-	H-C-	Ar-H
1	1-2	2-3	3.5	5	3	6-10
Ar-CH ₃	ket.	ald.	Ar-OH	R-OH	R-NH ₂	RCOOH
2.5	2.5	9.5	5-8	1-6	1-5	10-13

- H-NMR splitting patterns:** peak split by spin-spin interactions between adjacent H-atoms; "n" H's, give "n+1" peaks; example: $-\text{CH}_2-\text{CH}_3$ will have a quartet for the CH_2 and a triplet for the CH_3

- Quantifying H-NMR data:** The strength of the resonance signal, given by the area under the curve, is proportional to the number of H's producing the resonance. The relative peak-area gives the fraction of H-atoms in the compound associated with that peak. Temperature dependent NMR is used to explore fluxional distortions.

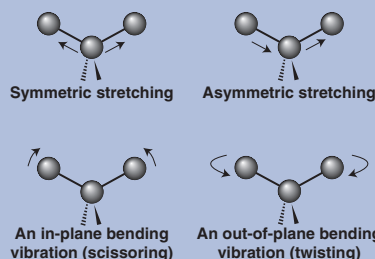
INFRARED (IR)



- IR excites vibrations which change the molecular dipole moment.
- Vibrational frequencies are characteristic of functional groups and bond-types; typically given in wavenumbers (ν , cm^{-1}), $1/\lambda(\text{cm})$.

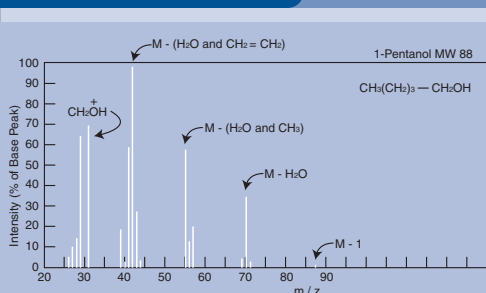
IR vibrational frequencies (wavenumber)

group	C=O	RO-H	R-OH	C-H	N-H
freq.	1700	3600	1200	2900	3400
group	H-Ar	>C=C<	-C $\equiv$ C-	RC $\equiv$ N	
freq.	3000	1650	2200	2250	



- Isotope effects:** isotopic substitution changes the reduced mass (with little effect on spring constant), shifting the vibrational frequencies

MASS SPECTROMETRY



- An electron-beam ionizes and fragments the molecules in a vacuum chamber. The molecular ions are sorted by mass/charge (M/z) using a magnetic field.
- The observed spectrum is " M/z vs. intensity."
- The fragmentation pattern gives the makeup of the molecule.
- Interpretation requires **isotope masses**, not **atomic weights**.

CHROMATOGRAPHY

- Solubility and surface-interactions separate a mixture.
- The mobile phase carries the sample, which interacts with the stationary phase.
- The greater the interaction between a sample component and stationary phase, the longer the material stays on the column, giving a separation over time.

paper chromatography: liquid-solvent carries sample along a paper strip

column chromatography: sample passes through a high-surface-area matrix

instrumental separation methods; HPLC (High Performance Liquid Chromatography): sample carried by a liquid mobile phase, interacts with a solid column

gas chromatography (GC): vaporized sample is carried by a flow of inert gas through a porous-packed solid or coated column

THERMODYNAMICS

The study of the heat and work associated with a physical or chemical process.

Key Thermodynamic Variables

Enthalpy (H):

ΔH = heat absorbed or produced by a process under constant pressure (normal lab conditions).

$\Delta H < 0$ for exothermic, $\Delta H > 0$ for endothermic

Enthalpies of Formation, ΔH_f° :

$$\Delta H = \sum \text{product } \Delta H_f^\circ - \sum \text{reactant } \Delta H_f^\circ$$

Entropy (S):

ΔS = change in thermodynamic disorder for a process

Standard Entropy, S° :

$$\Delta S = \sum \text{prod } S^\circ - \sum \text{react } S^\circ$$

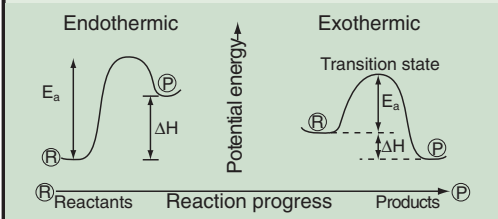
Gibbs Free Energy (G):

$\Delta G = \Delta H - T\Delta S$. ΔG is the capacity of the system to perform work. $\Delta G = 0$ at equilibrium, $\Delta G < 0$ for spontaneous (large K_{eq}), for $\Delta G > 0$, the reverse process is spontaneous.

Endergonic: $\Delta G > 0$; **Exergonic:** $\Delta G < 0$. $\Delta G = -RT \ln(K_{eq})$

Free energy of formation, ΔG_f° :

$$\Delta G = \sum \text{product } \Delta G_f^\circ - \sum \text{reactant } \Delta G_f^\circ$$



KINETICS: RATE OF CHEMICAL REACTION

For a generic reaction, $A+B \Rightarrow C$, the reaction rate is defined as the rate of producing C (or consuming A or B); the rate law describes the mathematical dependence of the rate on [A].

FIRST-ORDER:

Rate = $k_1[A]$

One species is involved in the rate determining step.

"ln [A] vs. time" is linear, the slope is the rate constant k_1 .

- Half-Life ($t_{1/2}$)** characterizes the process. [A] decays exponentially with time; $[A] = [A]_0 e^{-kt}$.

Examples: radioactive decay, unimolecular decomposition, SN1, E1 (carbocation), molecular rearrangement

SECOND ORDER:

Rate = $k_2[A]^2$ or $k_2[A][B]$

Two species in the rate determining step.

Examples: SN2, E2, acid-base

MULTIPLE-STEP REACTION:

Complicated rate-law; focus on rate determining step. The intermediate formed at this step can be modeled using **transition-state-theory**. The **steady-state approximation** works for reactions with unstable intermediates.

TEMPERATURE AND RATE CONSTANT (k)

Arrhenius Law: $k = A e^{-E_a/RT}$

- E_a : activation energy

- Plot of "ln(k) vs. $1/T$ " is linear;

slope is $-E_a/R$, intercept is $\ln(A)$

- T:** temperature in Kelvin (not $^\circ\text{C}$)!

- catalyst:** decreases E_a and accelerates the reaction

ORGANIC CHEMISTRY

REACTIONS

FEATURES OF AN ORGANIC REACTION

Mechanism: Describes the overall reaction using a series of simple steps.

Stoichiometry: Calculate reactant and product masses using the balanced equation and molar masses.

Kinetics: The study of the reaction rate and mechanism.

Theoretical yield: Mass of product given by a complete reaction;

$$\% \text{yield} = 100 \times (\text{product mass}) / (\text{theoretical yield})$$

Equilibrium: Reaction does not proceed to completion, instead, it reaches a balanced state of forward and reverse reactions.

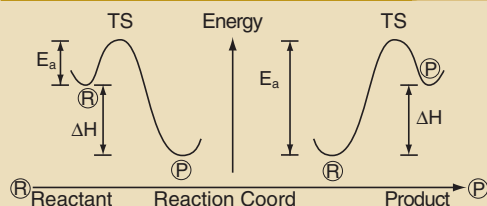
Major reaction types:

- Acid
- Base
- Oxidation Reduction
- Condensation
- Substitution (SN1, SN2)
- Ionic Reaction
- Elimination (E1, E2)
- Cyclization
- Hydrolysis
- Addition
- Radical reaction

Important named reactions:

- **Diels Alder:** form cyclic alkene
- **Friedel-Crafts:** add acyl or alkyl group
- **Grignard:** add alkyl or aryl group
- **Wolf-Kirschner, Clemmensen:** reduce ketone to alkane
- **Wittig:** convert aldehyde/ketone to alkene

KINETICS AND REACTION MECHANISM



Transition state (TS): Maximum on the reaction-coordinate curve: the least stable intermediate.

Activation Energy (E_a): Energy of the TS relative to the reactant. The change in enthalpy (ΔH) is < 0 for **exothermic**; > 0 for **endothermic**.

Hammond-Leffler postulate: The TS is more like the reactant or product that is closer in energy; *endothermic* TS is like the product, *exothermic* TS is like the reactant.

Kinetic vs. thermodynamic control: ΔG and ΔH describe Thermodynamic Stability.

- If ΔG is large and negative (**exergonic**), the product formation is likely controlled by "thermodynamics." Large K_{eq} corresponds to a large amount of product, relative to reactant.
- A large E_a may give rise to "kinetic" control; the energy of the TS controls the reaction, instead of the product-reactant thermodynamics.

Solvent effects: A solvent may stabilize an intermediate, decreasing E_a and increasing the rate of the reaction. Charged-complexes are stabilized by polar solvents.

ORGANIC ACID AND BASE

Acid:

- Electron-pair acceptor (Lewis acid)
- Proton donor (Bronsted-Lowry acid); example: carboxylic acid

Base:

- Electron-pair donor (Lewis base)
- Proton acceptor (Bronsted-Lowry base); example: *amine*

Factors enhancing acid strength (HA):

- Weaker H-A bond
- Greater electronegativity of "A"
- Inductive effect of substituent on "A" (electron withdrawal enhances transfer).
- More "s" character in hybrid orbital (s-orbital is lower in energy than p-orbital)
- Resonance stabilized conjugate base (A^-)

Factors enhancing base strength:

- Reverse of acid-strength guidelines

A base is a nucleophile; Electronic effects which shift electron density to the atom with the lone-pair increases base-strength.

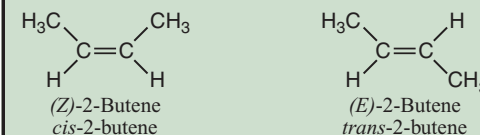
ALKENE $>C=C<$

Properties: Similar to alkane; non-polar, flammable

Nomenclature:

- Add -ene to prefix; Use # to denote C=C position
- **Isolated** C-C=C-C-C=C; **cumulative** -C=C=C-
- Polyunsaturated fatty acid: 2 or more C=C
- Allene: adjacent C=C=C
- Vinyl group: $H_2C=CH-$
- Methylene group: $H_2C=$
- Allyl group: $H_2C=CH-CH_2-$
- Vinyl halide: halide replaces -H on $>C=C<$
- Conjugated: alternate C-C and C=C (resonance)
- Alkadiene, 2 conjugated C=C; example: butadiene; s-cis and s-trans (rotate about C-C bond)
- Alkatratriene, 3 conjugated C=C
- Annulene: conjugated monocyclic compound; example: [6] annulene = benzene
- Aromatic cyclic ions: cyclopentadiene anion, cycloheptatriene cation (6 electrons)

Isomers: no free rotation of C=C



- E/Z; prioritize groups by atomic weight (Z - higher priority groups on the same side)
- For noncyclic: *cis* is less stable (steric hindrance). For cyclic, *cis* more stable.
- **Hofmann Rule:** Form the least-substituted alkene
- **Markovnikov Addition:** H adds to C with most H's
- **Zaitsev Elimination:** Form alkene with more substitution

Synthesis:

- Dehydrate alcohol (H^+ , heat) (elimination)
- Dehydrohalogenate haloalkane (base, heat)
- Dehalogenate vic dihalide (Zn, acetic acid)
- Hydrogenate alkyne:
 - syn, Z/*cis*-isomer (H_2 , P-2 catalyst)
 - anti, E/*trans*-isomer (Li, NH_3 , $-78^\circ C$)
- Wittig, aldehyde/ketone + phosphorous ylide

Reaction:

- Combustion (O_2)
- Hydrate to 2°/3° alcohol (H^+ , H_2O); 1° from ethene; can rearrange (Markovnikov)
- Hydrate to alcohol; hydroborate/oxidize ($THF/B_2H_6, H_2O_2/OH^-$) (syn, anti-Markovnikov)
- Oxymmercuration-demercuration to alcohol
- Hydrohalogenate (HX) (Markovnikov)
- Halogenate (Br_2/Cl_2), vic dihaloalkane (X_2, CCl_4 ; anti)
- Halohydrin (X_2, H_2O ; anti-addition)
- Hydroxylate to form a 1,2-diol ($KMnO_4$, cold OH^- ; syn addition)
- Oxidize to carboxylic acid ($KMnO_4$, hot OH^-)
- Ozonolyse to ketone (O_3 ; Zn, H_2O)
- Hydrogenate to alkane (Pt, H_2 ; syn-addition)
- Free radical polymerization
- Alkadiene Reaction
 - allylic halogenation (Cl_2 , heat)
- Diels-Alder: cycloalkene from diene + alkene/alkyne

ALKANE



Properties:

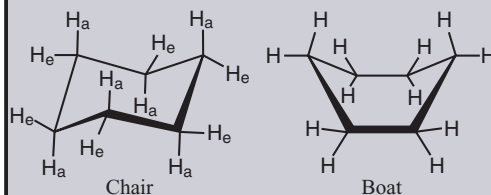
- Hydrocarbon
- Weak intermolecular forces
- Non-cyclic: general formula C_nH_{2n+2}
- Tetrahedral C-C-C (109°)

Nomenclature:

- Add "-ane" to prefix
- Locate substituent by position #
- Haloalkane: substitute halide for -H

Cycloalkane: (C_nH_{2n})

- Bicyclic - two fused or bridged rings
- $n = 3$: **cyclopropane**: (highly strained)
- $n = 4$: **cyclobutane**: (some flexibility)
- $n = 5$: **cyclopentane**: (slight puckering)
- $n = 6$: **cyclohexane**: *chair* - stable conformer; *boat* - less stable; *Axial* position: "perpendicular" to ring; *Equatorial* position: in ring "plane" (see H_a and H_e in chair diagram below)
- *Cis* - two substituents in *up* position
- *Trans* - *one* up and *one* down



Synthesis:

- Hydrogenate alkene or alkyne (H_2 , Pt catalyst)
- Free-radical reaction of alkene
- Reduce haloalkane (Zn, H^+)
- Friedel-Crafts alkylation

Reaction:

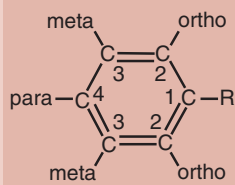
- Combustion: $alkane + O_2 \Rightarrow CO_2 + H_2O$
- Halogenation to haloalkane (Cl_2/Br_2 , light or heat)

BENZENE/ARENE

Properties: insoluble in water, miscible with non-polar organic solvents.

Nomenclature:

- Aromatic** (or arene): Denote substituent using group name and ring position; ortho (1,2), meta (1,3), para (1,4);
- examples: benzene C_6H_6 ; phenol, Ar-OH (carbolic acid, hydroxybenzene, benzenol); aniline Ar-NH₂; toluene, Ar-Me (methyl benzene); xylene, dimethyl benzene
- Fused rings:** naphthalene, $C_{10}H_8$ (2 edge-sharing rings)
- Aryl or Phenyl group:** Ar- (remove H from benzene)
- Aryl halide:** halogen replaces an H atom; Ar-X
- Alkenyl benzene:** Ar-C=C<
- Benzyl:** Ar-CH₂-



Synthesis: Dehydrogenate cyclohexane (sulfur+ heat)

General Reaction:

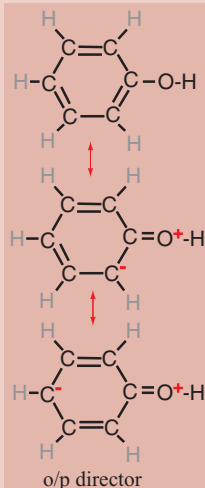
- Combustion (similar to alkane)
- Birch reduction => 1,4 cyclohexadiene (Na, NH₃, EtOH)
- Hydrogenate to cyclohexane (H₂, Pt)

Electrophilic substitution:

- Alkylation: Ar-R (*Friedels-Craft*, RCl, AlCl₃)
- Nitration: Ar-NO₂ (HNO₃, H₂SO₄)
- Halogenation: Ar-Br (Br₂, FeBr₃)
- Ar-Cl (Cl₂, FeCl₃) Ar-I (I₂, HNO₃)
- Acylation: Ar-CR=O (RCOCl, AlCl₃)
- Sulfonation: Ar-SO₃H (SO₃, H₂SO₄)

Reactivity of substituted benzene:

- A substituent alters the ring electronic structure.
- Activating group:** More reactive than benzene; add electrons to the ring, stabilize the arenium cation
- Deactivating group:** Less reactive; pull electrons from the ring, destabilize the arenium cation
- Ortho/para-director:**
 - substituent tends to activate the ring (except for -X); electron density donated to ring creates “-” center on o/p sites, o/p isomers are preferred
 - examples: -NR₂, -OH, -R, -OR, -X (halogen)
- Meta-director:**
 - substituent tends to deactivate the ring; electron density withdrawn from the ring creating “+” center on o/p site, m- preferred reaction site.
 - examples: -NO₂, -CN, -COOH, -SO₃H, -COOR, -CHO, -CRO

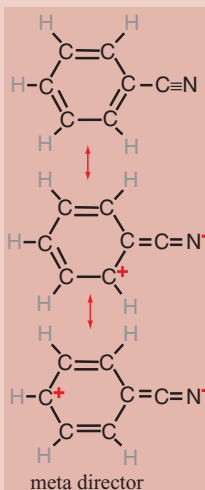


Reactivity of di-substituted benzene:

- Directing effects may be cooperative; e.g. “o/p” plus “m” at 1,4 positions
- Otherwise: consider steric effects; activating group tends to dominate deactivating group.

Reaction of alkyl substituted benzene:

- Toluene to benzoic acid: (KMnO₄, OH⁻, heat, H⁺)
- Chlorinate -Me of Toluene (Cl₂)



ALKYNE -C≡C-

Properties:

- Hydrocarbon, at least 1 C ≡ C triple bond
- Properties similar to alkane or alkene
- Linear R'-C≡C-R'

Nomenclature:

- Add -yne to prefix
- Number denotes position of triple bond; example: ethyne (acetylene) C₂H₂

Synthesis:

- CaC₂ + H₂O => Ca(OH)₂ + C₂H₂
- Dehydrohalogenate vic-haloalkene (NaNH₂, liq NH₃)
- Alkylate terminal alkyne (NaNH₂, liq NH₃; R-X)

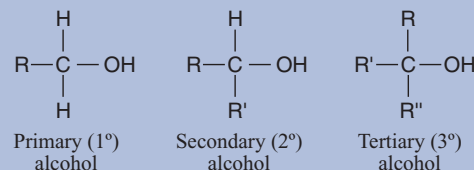
Reaction:

- Addition: hydrogenate to alkane (H₂, Pt or Ni)
 - syn to cis/Z alkene (H₂/Ni₂B P-2 catalyst)
 - anti to trans/E alkene (Li, Liq NH₃)
- haloalkene to gem-dihalide (HX) (Markovnikov)
- halogenate to haloalkene or haloalkane (X₂)
- Ozonolyze to carboxylic acid (O₃, H₂O)
- Oxidize to carboxylic acid (KMnO₄, OH⁻; H⁺)

ALCOHOL R-OH

Properties:

- Low molecular weight are water-soluble
- H-bonding, polar
- RO-H acidic proton
- Resonance stabilized ArO⁻ or RO⁻



Nomenclature:

- Prefix + “anol”; example: methanol Me-OH (methyl alcohol)
- Cyanohydrin:** -OH and -CN
- Halohydrin: -OH and halogen
- Diol or glycol (two -OH); **gem-diol:** 1,1 diol; **vic-diol:** 1,2 diol

Synthesis:

- Hydrate alkene (H₂O, H⁺)
- Hydroborate/oxidize alkene (THF:BH₃; H₂O₂, OH⁻)
- Hydrogenate aldehyde (H₂/Ni or Pt catalyst)
- Hydrolyze 1o alkyl halide (water, OH⁻)
- Reduce aldehyde, ester, ketone or carboxylic acid
- Ethanol: Ferment sugar or starch
- Methanol: CO + H₂, catalyst; Pyrolyze cellulose
- Hydrolyze ester (water, acid)
- Dehydrate ether (H₂SO₄, low heat)
- Grignard (RMgX): formaldehyde => 1° alcohol; aldehyde => 2° alcohol; ketone => 3° alcohol
- Synthesis of Glycol from Ketone/aldehyde: (HIO₄ or Pb(OAc)₄; H₂SO₄ + heat)
 - oxidize alkene: (KMnO₄: cis) (H₂O₂, formic acid: trans)
- hydrolyze epoxide (H₂O, H₂SO₄)

Reaction:

- Oxidize 1° to aldehyde (CuO, heat) or 2° to ketone (KMnO₄, H⁺)
- Oxidize 1° to carboxylic acid (KMnO₄, H⁺)
- Dehydrate to alkene; Zaitsev's rule; rate 3° > 2° > 1° (hot H₂SO₄, or Al₂O₃)
- Dehydrate to ether (H₂SO₄, lower temperature)
- Oxidize to ketone (2° alcohol) (H₂CrO₂)
- Form haloalkane (HX; substitution)

AROMATIC ALCOHOL Ar-OH

Properties:

- The most common is Phenol, Ar-OH
- Acidic hydrogen, ArO-H; pK_a=9.9
- Ring substituent alters acidity
- Benzendiols, HO-Ar-OH; para, hydroquinone; ortho, catechol; meta, resorcinol

Reaction of Phenol:

- Electrophilic substitution: o-p director
- Hydrogenate to cyclohexane (H₂, catalyst)
- Form ester (acid anhydride or acid chloride)

Synthesis of Phenol:

- Electrophilic aromatic substitution
- Williamson reaction, phenyl ether (NaOH, RCH₂X)
- Arenediazonium salt intermediate: Ar-NH₂ + HONO => Ar-N₂⁺ + Cu₂O, H₂O => Ar-OH
- Benzene + propene => cumene; oxidation/acid => phenol + acetone
- Aryl halide (Ar-X) + NaOH, heat and acid
- Ar-OR + HI/HBr, heat

HALOALKANE/ALKENE/ARENE R-X

Nomenclature:

- Halogen (X = fluorine, chlorine, bromine or iodine) replaces -H on hydrocarbon group
- Denote halogen in the name; example: Chloromethane: Cl-Me; chlorobenzene Ar-Cl

Synthesis: alcohol (ROH) + HX

Reaction:

- Dehydrohalogenate to alkene (often rearranges)
- Hydrolyze 1° alkyl halide to alcohol (RX + OH⁻)

HALOHYDRIN X-R-R'-OH

Synthesis: Alkene + X₂, H₂O

Reaction:

- Halohydrin + ROH => β hydroxy ether
- Halohydrin + RNH₂ => β hydroxy amine
- Halohydrin + RSH => β hydroxy sulfide

ETHER R''-O-R'

Properties:

- Polar, hydrogen bonding
- Oxygen lone-pair is a nucleophile
- Flammable liquid

Nomenclature:

- R''-O-R', “R R' ether” or “alkoxy alkane”;
- Example: diethyl ether, common solvent: Et-O-Et
- Alkoxy** group, -OR (O-Me, methoxy; O-Et, ethoxy)
- Oxa-: substitute an -O- for a -CH₂-
- Cyclic ether: tetrahydrofuran (THF)
- Epoxide** or **oxirane:** 3-member ring
- Dioxane:** cyclic double ether
- Peroxide:** R-O-O-R' ; -O-O- single-bond

Synthesis:

- Williamson synthesis (R'I + NaOR)
- Dehydrate 1° alcohol (H₂SO₄, heat)
- Epoxidation: alkene + peroxyacid
- Halohydrin + ROH => hydroxy ether

Reaction:

- Hydrolyze to alcohol (H⁺ or OH⁻)
- Autoxidize to peroxide (oxygen in air); **EXPLOSIVE HAZARD!**

Epoxide reaction:

- Hydrolyze 1,2 glycol (acid, H⁺)
- Hydrolyze to 1,2 glycol (base, OH⁻ or OR⁻)
- Grignard + epoxide => 1° alcohol

ALDEHYDE & KETONE >C=O

Properties:

- Polar $>C^+=O$; low molecular weight are water-soluble
- Main chemical difference: ketone is harder to oxidize than aldehyde.

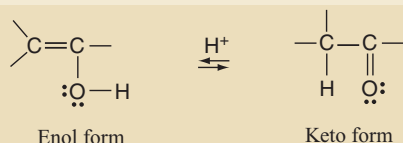
Aldehyde nomenclature (RCHO):

- Prefix + "anal";
- Example: HCHO, methanal (formaldehyde); MeCHO, ethanal (acetaldehyde); Ar-CHO, benzaldehyde

Ketone nomenclature (RR'CO):

- Prefix + "anone," also "R,R' ketone";
- Example: 2-propanone (acetone or dimethyl ketone);
- Diketone: 2 $>C=O$ groups
- Acyl group: $RC=O$ or $Ar-C=O$
- Ketene: $C=C=O$
- Ketal: $RR'C(OR)(OR)$;
- Acetal: $RHC(OR)(OR)$
- Hemiacetal: $RHC(OH)(OR)$
- Diketone: $R'-CO-CH_2-CO-R$

Keto-enol tautomerism:



- Nucleophile attacks $>C=C<$ of enol-form
- Acidic α -H, $-CH^*-CHO$ can form resonance stabilized carbanion (especially for diketone).
- Racemization via keto-enol: chiral ketone $\Rightarrow$ achiral enol $\Rightarrow$ achiral ketone

Synthesis:

- Oxidize alcohol: aldehyde from 1° (Cu, heat); ketone from 2° (H_2CrO_4)
- *Grignard*: nitrile (RCN) + $R'MgX \Rightarrow RCR'O$
- Reduce RCO_2R' (i-Bu₂AlH)
- Reduce RCN (i-Bu₂AlH)
- Ozonolyze alkene (O_3 , H_2O_2)
- *Friedel-Craft* acylation: $Ar-H + RCOCl (AlCl_3)$

General Reaction:

- *Wittig*, form alkene (phosphorous ylide)
- Form Oxime ($>C=N-OH$) (hydroxyl amine)
- Reduce to alcohol (Metal hydride, $LiAlH_4$)
- *Wolff-Kishner*: $>C=O$ to $>CH_2$ (N_2H_4 , base, heat)
- *Clemmenson* reduction, $>C=O$ to $>CH_2$ ($Zn(Hg)$, HCl)
- Hydrogenate to ROH (H_2 , metal; $NaBH_4$, H^+ ; $LiAlH_4$, H^+)
- Oxidize to RCOOH (peroxyacid)
- Form cyanohydrin (HCN)
- Form imine ($>C=N-R$) (1° amine)
- Aldol condensation, $>CH=O + COOH \Rightarrow >C=C-CH=O$
- Nucleophilic attack: $RCHO + H-Nu \Rightarrow R-C(OH)-Nu$
- Hemiacetal/ketal formation: $ROH + R'_2C=O \Rightarrow R'_2C(OH)(OR)$
- Formation of acetal ($R'OH$, HCl)
- Reductive amination: aldehyde or ketone $\Rightarrow$ amine (amine or ammonia, H_2 , Rh)

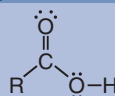
Specific Reaction:

- Acetaldehyde to gem-diol (H_2O , H^+ or OH^- catalyst)
- Synthesis of acetaldehyde (C_2H_2 , Hg^{2+} , H^+ , H_2O)
- Oxidize aldehyde to RCOOH: Ag_2O, OH^- or $Ag(NH_3)_2^+$; *Tollen's* reagent, ketone is not oxidized
- Haloform, methyl ketone (X_2 , OH^-)
- Halogenate -H of ketone (X_2 , H^+ or OH^-)

CARBOXYLIC ACID

Properties:

- Organic acid, resonance stabilizes dissociation
- Soluble in water; H-bonding, acid strength given by pKa



Nomenclature:

- Prefix + "oic acid";
- Examples: HCOOH, **methanoic acid** (formic acid); Me-COOH, **ethanoic acid** (acetic acid); Ar-COOH, **benzoic acid** (benzenecarboxylic acid); **oxalic acid** (dicarboxylic acid, HOOC-COOH); **malonic acid** (HOOC-CH₂-COOH)
- **Fatty acid**, "R" long hydrocarbon (aliphatic) chain

Derivatives:

- Ester
- Amide
- Acid anhydride: $RCO-O-CO-R$
- Peroxyacid: $R-CO_3H$
- Acyl chloride
- Amino acid

Synthesis:

- Oxidize 1° alcohol ($K_2Cr_2O_7$, OH^-)
- Oxidize aldehyde (Ag_2O , H^+)
- Oxidize alkene ($KMnO_4$, OH^- , heat, H^+)
- Ozonolyze alkene (O_3 , H_2O_2)
- Hydrolyze nitrile or acyl chloride (H^+ , H_2O)
- Acid anhydride + water
- Grignard and carbonylation ($RMgX + CO_2$, H^+)
- Benzoic acid: oxidize $1^\circ/2^\circ$ alkylbenzene ($KMnO_4, OH^-, heat, H^+$)
- From methyl ketone ($Ar-CO-CH_3$) (X_2 , OH^- , H^+)

Reaction:

- Form acyl chloride ($SOCl_2$, PCl_3 or PCl_5)
- Reduce to alcohol ($LiAlH_4$)
- Neutralize with a base, form a salt
- Esterification: ($R'OH$, H^+)
- Reduce to ketone ($Ba(OH)_2$, heat)
- Decarboxylate keto acid to ketone (heat)
- α halo acid: (X_2 , P, H_2O): HVZ (Hell-Volhard-Zelinski)
- α hydroxy acid from halo acid (OH^- , H^+)
- α amino acid from halo acid (NH_3 or amine)

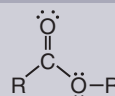
Carbonic acid and derivatives:

- Carbonic acid: H_2CO_3 or $HO-CO-OH$
- Carbonyl dichloride (phosgene), $Cl-CO-Cl$; toxic gas
- Phosgene + EtOH $\Rightarrow$ diethylcarbonate, $EtO-CO-OEt$
- Phosgene + $NH_3 \Rightarrow H_2N-CO-NH_2$ (urea)
- Phosgene + ROH $\Rightarrow RO-CO-Cl$ (alkyl chloroformate)
- $RO-CO-Cl + RNH_2 \Rightarrow RO-CO-NHR$ (urethane, carbamates)

ESTER

Properties:

- Derive from carboxylic acid; polar, weak H-bonding; pleasant or fruity odor



Nomenclature:

- Denote "alcohol" component with "-yl" suffix, acid with "-oate" or "-ate" suffix.
- Examples: Me-CO-O-Eth, ethyl acetate (ethanol+acetic acid);
- Lactone: cyclic ester

Synthesis:

- Esterification: $ROH + R'COOH \Rightarrow R'COOR$ (acid)
- Acid chloride ($RCOCl$) + $R'OH$
- $R-CN + R'OH (H^+)$
- Acid anhydride + alcohol $\Rightarrow$ ester + carboxylic acid
- Aromatic ester: phenol + carboxylic anhydride
- β -keto ester: Claisen condensation from ethyl acetate ($NaOEt$, HCl)
- Transesterification: $R'COOR + R''OH \Rightarrow R'COOR'' + ROH (H^+, heat)$

ESTER continued

Reaction:

- Acid-catalyzed hydrolysis
- Saponification: base-catalyzed hydrolysis
- Three fatty acids + ethylene glycol $\Rightarrow$ triglyceride
- Grignard to 3° alcohol ($R''MgX + R-COOR'$)
- Reduce to 1° alcohol (H_2 , Ni)
- Form amide ($RCOOR + 1^\circ/2^\circ$ amine)
- Pyrolyze to alkene and carboxylic acid

Lactone: Cyclic ester

- Intramolecular esterification of δ -hydroxy acid (H^+)
- Hydrolyze δ/γ lactone to δ/γ hydroxy acid (OH^-, H^+)

AMINE

RR'R''N

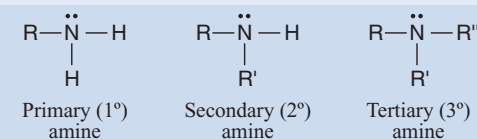
Properties:

- Substituted ammonia; polar, water soluble; $>N-H$ forms H-bonds
- **Organic base**: strength denoted by pKb
- **Structure**: distorted pyramid (AX_3E)

Nomenclature:

- "R1 R2 R3 amine"
- Example: Me-NH₂, methyl amine; Ar-NH₂, phenylamine (aniline, amino benzene)

Types of amines:



- Quaternary ammonium salt (4°) $NR'R'R'R^+$ cation (no lone-pair)

Synthesis:

- **1°**: amine haloalkane: $RCH_2X + NH_3$
 - reduce nitrile, RCN ($LiAlH_2$) or (H_2 , Ni)
 - reduce nitroalkane, RNO_2 ($LiAlH_4$)
 - reduce oxime (Na, EtOH)
 - from aldehyde/ketone (NH_3 , H^+)
- **2°**: haloalkane + 1° amine
aldehyde/ketone + $R'NH_2 (H^+)$
- **3°**: haloalkane + 2° amine
reduce amide ($LiAlH_4$, H_2O)
aldehyde/ketone + $R'R''NH (H^+)$
- Aromatic Amine: $Ar-NO_2 \Rightarrow Ar-NH_2$ (H_2 , catalyst; Fe, HCl, OH^-)

Reaction of amine:

- React as a base: amine + $H^+ \Rightarrow R_3NH^+$
- Nucleophilic N lone-pair
- Amine + sulfonyl chloride $\Rightarrow$ sulfonamide
- amide formation: $1^\circ + R'COCl \Rightarrow R'CO-NHR$
- $1^\circ + CH_3COOH \Rightarrow R-NO_2$
- amide formation: $2^\circ + RCOCl$
- **Cope Elimination**: oxidize 3° amine to tertiary ammonium oxide ($R_3N^+-O^-$), heat produces $RHC=CH_2$
- Ar-NH₂: o-p director, electrophilic aromatic substitution
- Ar-NH₂: nucleophilic aromatic substitution:
 - Step 1: $Ar-NH_2 + cold\ nitrous\ acid \Rightarrow Ar-N_2^+$ (diazonium salt, unstable)
 - Step 2: Depends on substitution:
 - + Cu_2O , Cu^{2+} , $H_2O \Rightarrow Ar-OH$
 - + $CuCl \Rightarrow Ar-Cl$
 - + $CuCN \Rightarrow Ar-CN$
 - + $H_3PO_2 \Rightarrow Ar-H$

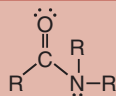
Hofmann elimination:

- Quaternary ammonium hydroxide $\Rightarrow$ alkene (heat)

AMIDE

Nomenclature:

- Example: Me-CO-NH₂, acetamide
- Cyclic amide (lactam): N of amide forms ring with β, γ or δ carbon;
- β forms 4 membered ring; γ forms a 5 membered ring, δ form a 6 membered ring.
- Observed in amino acids



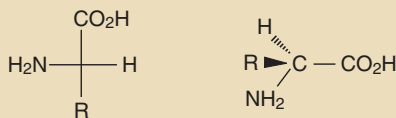
Synthesis:

- Nitrile hydrolysis (R-CN + H₂O, conc. H₂SO₄)
- Acyl chloride + 1°/2° amine or ammonia
- Pyrolysis of ammonium salt + RCOOH
- Ammonolysis of ester: 1° or 2° amine + ester
- Polyamide => polypeptide => protein

Reaction:

- Reduce to amine (LiAlH₄)
- Hydrolyze to acid (H₂O, H⁺ or OH⁻)
- Dehydrate to nitrile, RCN (P₄O₁₀, heat)
- *Hofmann* Reaction: Form 1° amine (NaOBr)
- *Grignard* (R⁻MgX) to ketone, R-CO-R⁻
- Form aldehyde and 2° amine (LiAlH₄, (OEt)₂)
- Nucleophilic substitution; Form R-CO-Nu + amine

AMINO ACID



Properties:

- Basic (-NH₂) and acidic (-COOH) functionality
- Chiral isomers
- **Zwitterion**: self-ionization of amino acid to produce COO⁻ and -NH₃⁺
- Isoelectric point, pH which produces equal + and - charges

Nomenclature:

- Common name based on "R" group; examples: glycine (-H), alanine (-CH₃)

Synthesis:

- Gabriel synthesis: RCH₂COOH + Br₂, PCl₃, NH₃

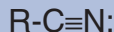
Reaction:

- Lactam formation (cyclic amide)
- Polypeptide formation (peptide bond); dehydration: R-NH₂ and HO-R' moieties
- Protein, amino acid polymer

OTHER NITROGEN-COMPOUNDS

Nitrile:

example: H₃C-CN; methane nitrile



Synthesis:

- Haloalkane + NaCN
- Aldehyde/ketone => cyanohydrin (CN⁻, H⁺)
- Dehydrate amide (P₄O₁₀, heat)

Reaction:

- Hydrolyze to carboxylic acid (acid, heat)
- Hydrolyze to carboxylate (base, heat)
- Reduce to 1° amine (Raney Ni; LiAlH₄)
- Form aldehyde (DIBAL-H (i-Bu)₂AlH, H₂O)
- Form ketone (*Grignard* reagent or R-Li, H⁺)

Imine:



Synthesis: Aldehyde/ketone + 1° amine (H⁺)

Reaction: Intermediate in amination of aldehyde/ketone

NITROGEN *continued*

Imide: R-CO-NH-CO-R'

Synthesis:

- Dehydration, amide + carboxylic acid

Oxime: >C=NOH

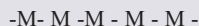
Synthesis:

- aldehyde/ketone + hydroxylamine

Reaction:

- oxime to 1° amine (Na, EtOH)

ORGANIC POLYMER



Monomers (M) bond to form a high molecular weight compound.

Factors which influence properties: chain length, branching vs. linear, nature of the monomer, density, interchain bonds, hydrophobic and hydrophilic interactions.

Examples:

- PE (polyethylene)
- PS (polystyrene)
- HDPE (high density polyethylene)
- LDPE (low density polyethylene)
- PET (polyethylene terephthalate)

Synthesis:

- Free-radical synthesis: ethylene => PE; styrene => PS (radical initiation)
- Condensation:
 - HO-R-OH+HO-R'-OH => HO-R-R'-OH + H₂O
 - Example: ethylene glycol and terephthalic acid => PET

Reaction:

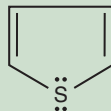
- Hydrolysis of polymer (reverse of condensation)
- Cross-link adjacent polymer chains or segments

SULFUR CHEMISTRY

Sulphur Compounds

- Thiol: R-SH
- Sulfide or Thioether: R-S-R'
- Disulfide: R-S-S-R'
- Thiol ester: R-CO-SR'
- Sulfoxide: R-S-OR'
- Sulfone: R-SO₂-R'
- Thiophenol: Ar-SH
- Thioketone: R-CS-R'
- Sulfonic acid: R-SO₃H
- Sulfinic Acid: R-SO₂H
- Hydrogen sulfate: R-OSO₃H

Thiophene, Heterocyclic sulphur compound



Synthesis:

- Thiol: From alkyl bromide/iodide (KOH, H₂S)
- Thiol: RCH₂X + NaSH => RCH₂SH (EtOH, heat)
- Thiol ester: Acyl chloride + thiol
- Alkyl hydrogen sulfate (Alkene + cold conc. H₂SO₄)
- Thiol: Alkene + H₂S (H₂SO₄, heat) (Markovnikoff addition)
- Thiol: Alkene+H₂S (peroxide or UV) (Anti-Markovnikoff addition)

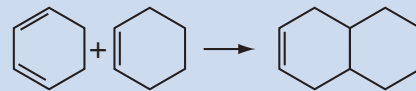
Reaction:

- Form sulfide from thiol (NaOH, R'CH₂X)
- Form disulfide from thiol (I₂ or H₂O₂)
- Oxidize thiol to form sulfonic acid, RSO₃H, (HNO₃)
- Desulfurization of thiol to alkane (H₂, Ni)
- Sulfonate benzene (SO₃, conc. H₂SO₄)

CYCLIZATION: SYNTHESIS OF A CYCLIC COMPOUND

Synthesis:

- Diels-Alder: diene + dienophile + heat => adduct



Diene

Dienophile

Adduct

- Freund-Gustavson: 3-membered ring from 1,3 dihalide (EtOH, Zn, heat)
- [2,2] cycloaddition of alkenes giving cyclobutane adduct (two alkenes, photochemical reaction)

Reaction of cyclic compound:

- Retro-Diels-Alder: thermally decompose cycloalkene
- Reduce aromatic to symmetric 1,4 cycloalkene (Li or Na, EtOH, Liq NH₃) (Birch)
- Small ring is strained, may decompose to linear chain
- Epoxide ring opening reaction

METAL REACTION

Organometallic:

- Carbon atom bonded to a metal atom
- Types of bonding:
 - ionic bond, Na, K; R⁻-M⁺
 - partial covalent, Mg, Li; R electrophilic character
 - covalent, Pb, Sn, Hg; R-M

Grignard reagent:

- Strong base gives R electrophilic character:
 - Li + R-Br => R-Li
 - RX + Mg => RMgX
 - ArX + Mg => ArMgX

Organoborane:

- Boron hydride, B_nH_m example: diborane, B₂H₆
- **Synthesis:**
 - Hydroboration: Alkene + Boron hydride syn addition
- **Reaction:**
 - Organoborane => alcohol (H₂O₂/OH⁻)
 - R-B< => R-H (acetic acid; addition of H)

Organolithium: R-Li

Synthesis:

- Li + haloalkane (R-X or Ar-X) (cold, Et₂O)

Organomagnesium: RMgX or ArMgX

- *Grignard*: RX + Mg (Et₂O); R behaves as R⁻

Organocopper: R-Cu

- Add R- to C=C of unsaturated carbonyl

Organolead/mercury:

- Stable compound, **VOLATILE AND TOXIC**
- Tetraethyl lead (anti-knock agent in gasoline)